NYC Yellow Taxi Trip Records

Insights and Trends

Aiden Pham Dang Khoa
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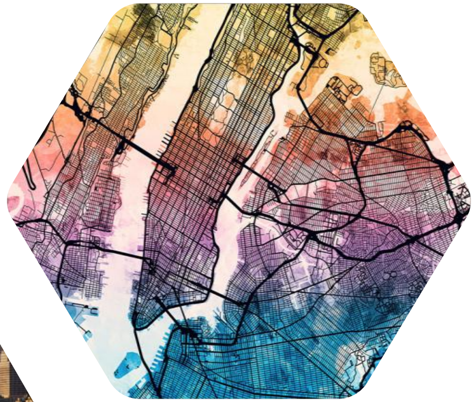
OUTLINE

1. Introduction
2. Data quality
3. Data cleaning
4. Feature engineering
5. Data analysis
6. Conclusions



OUTLINE

1. Introduction

1.1 Dataset

1.2 Research questions

1.3 Business significance

1.4 Methods

2. Data quality

3. Data cleaning

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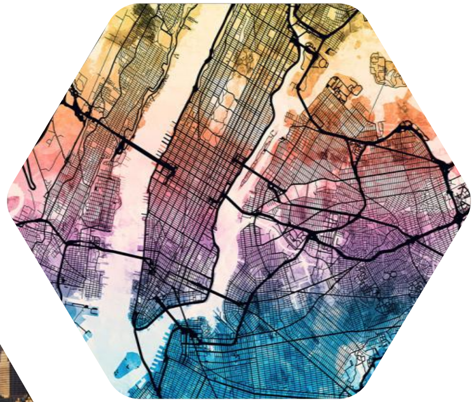


1. INTRODUCTION

1.1 Dataset

- **~305,000** NYC Yellow Taxi trips from **January 2018 to January 2023**
- 2018–2022: **~60,000** trips per year; 2023: January only (~**5,000** trips)
- Data appears sample-capped at ~5,000 trips/month - ratio metrics are reliable, absolute volumes are not.
- **19 columns**:
  - Pickup & dropoff time
  - Pickup & dropoff zone ID
  - Trip distance
  - Other dimensions (vendor, rate code, payment type, etc.)
  - Financial columns (fare, tax, surcharge, total, etc.)

```
RangeIndex: 304978 entries, 0 to 304977
Data columns (total 19 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   VendorID               304978 non-null  category
 1   tpep_pickup_datetime   304978 non-null  datetime64[ns]
 2   tpep_dropoff_datetime  304978 non-null  datetime64[ns]
 3   passenger_count        295465 non-null  category
 4   trip_distance          304978 non-null  float64
 5   RatecodeID             295465 non-null  category
 6   store_and_fwd_flag     295465 non-null  category
 7   PULocationID           304978 non-null  category
 8   DOLocationID           304978 non-null  category
 9   payment_type           304978 non-null  category
 10  fare_amount            304978 non-null  float64
 11  extra                  304978 non-null  float64
 12  mta_tax                304978 non-null  float64
 13  tip_amount             304978 non-null  float64
 14  tolls_amount           304978 non-null  float64
 15  improvement_surcharge  304978 non-null  float64
 16  total_amount           304978 non-null  float64
 17  congestion_surcharge   232346 non-null  float64
 18  airport_fee            106217 non-null  float64
dtypes: category(7), datetime64[ns](2), float64(10)
```

1. INTRODUCTION

1.2 Research Questions

Analyzed in section	Topics
5.1 Overview	Citywide demand, revenue, payment, and efficiency patterns
5.2 Time	Demand and efficiency by month, weekday, and hour
5.3 Space	Demand and efficiency hotspots by zone and borough
5.4 Vendor	Vendor performance comparison
5.5 Rate code	Rate-code segment performance comparison
5.6 Tip	Tip behavior by time and location
5.7 Fare	Relationship between fare, distance, and duration

1. INTRODUCTION

1.3 Business significance

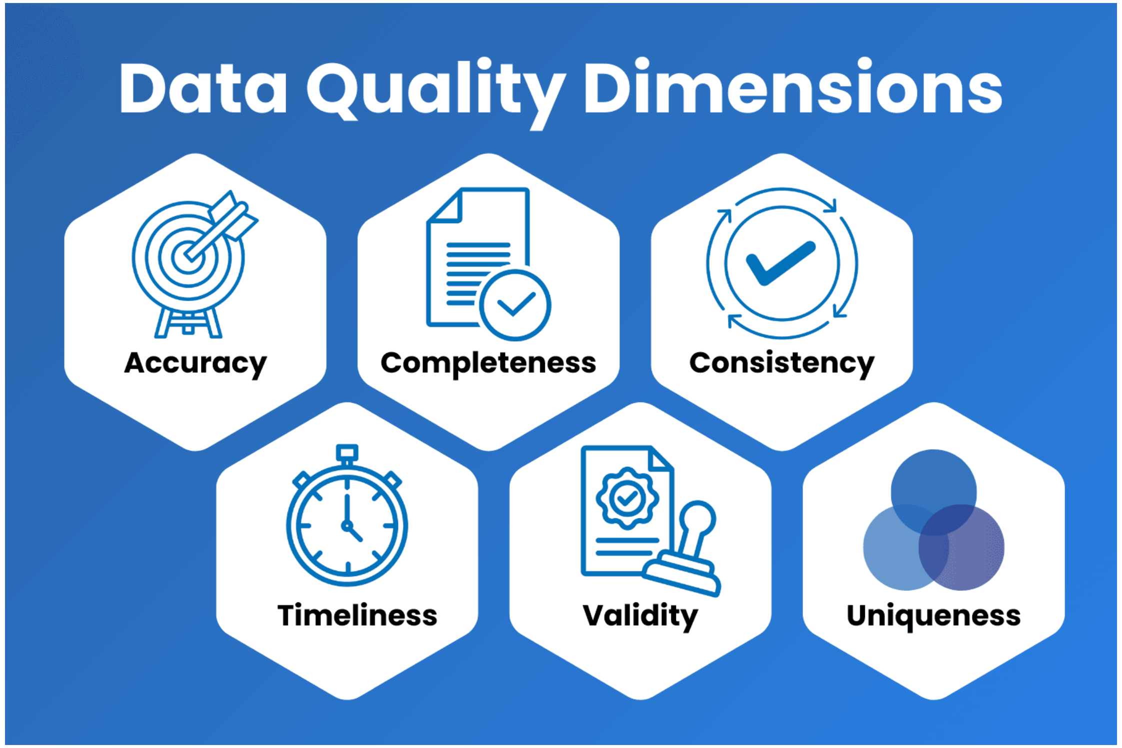
- Provide **clear, actionable KPIs**.
- Support **smarter fleet allocation.**
- Optimize **revenue management**.
- Enable more **targeted service strategy**.
- Improve **data quality**.

1. INTRODUCTION

1.4 Method

Data preparation → Data Quality Check → Data Cleaning → Feature Engineering → Analysis → Conclusions

2. DATA QUALITY

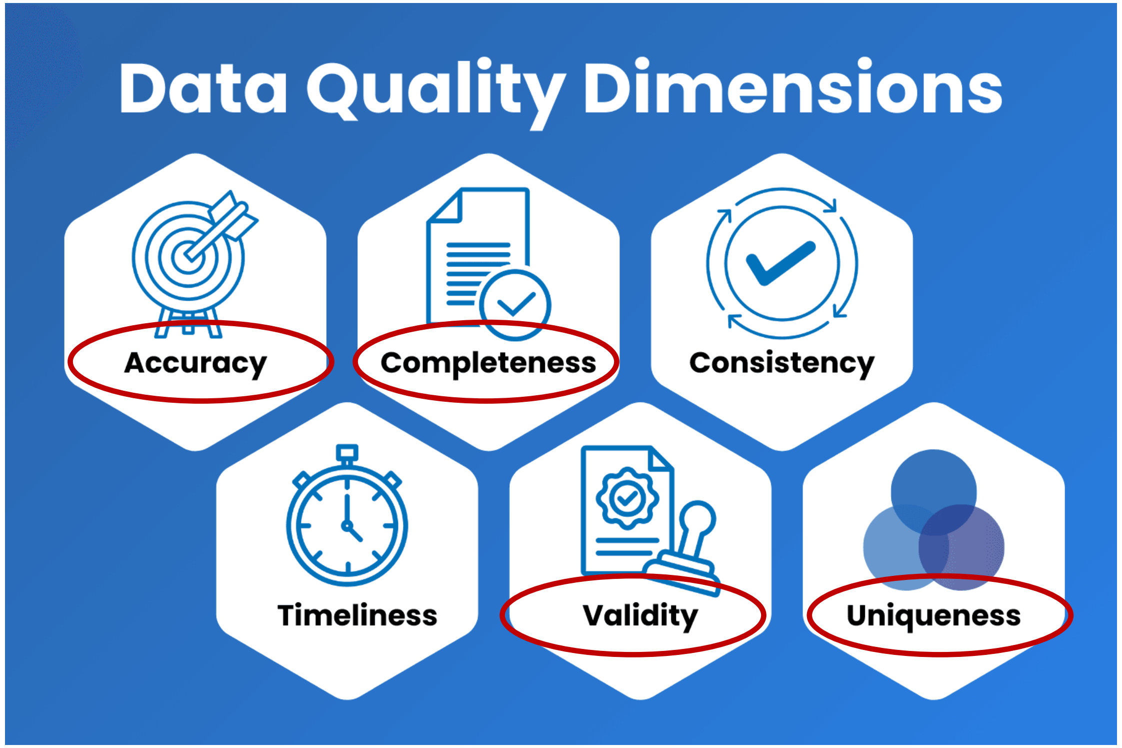


In the scope of this assignment, I will focus on Completeness, Uniqueness, Validity and Accuracy.

The other 2 dimensions are not relevant in this assignment:

- Timeliness (how up-to-date data are): this is a static dataset
- Consistency (data are consistent among tables, systems): there is only 1 dataset

2. DATA QUALITY



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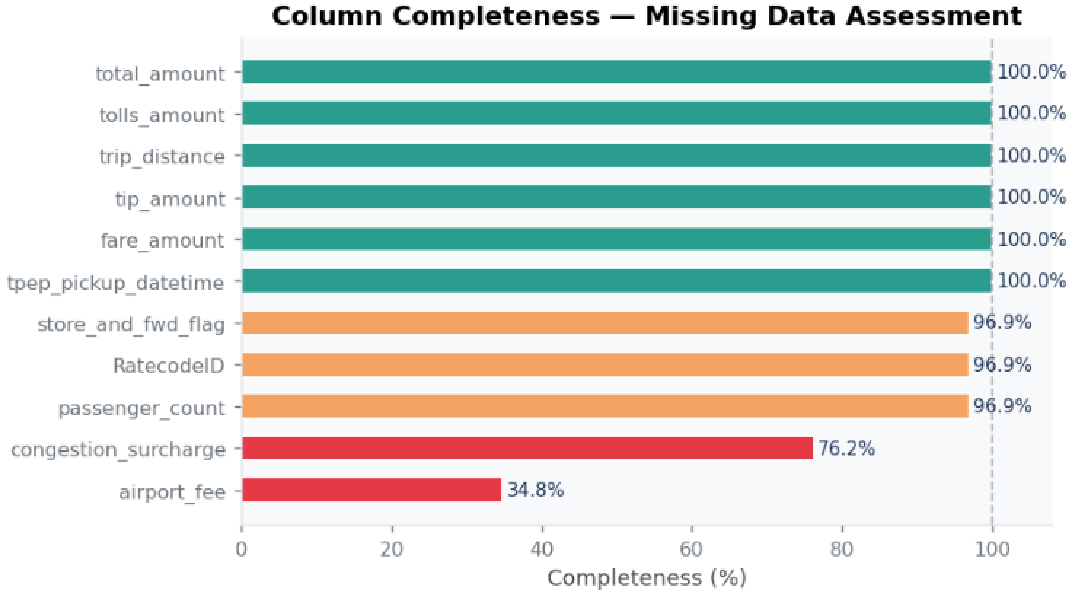
2. DATA QUALITY

2.1 Completeness

- ~3% of rows had NULL values in **passenger_count**, **RatecodeID**, and **store_and_fwd_flag** - all three missing together, suggesting a systematic data collection issue
- **congestion_surcharge** and **airport_fee** have NULLs that are not substitutes for zero - retained as missing



2.2 Uniqueness

- 3 pairs of duplicate trips identified based on identical pickup/dropoff time and distance
- Each pair showed a specific issue: NULL metadata (2 pairs) or negative financial values (1 pair)

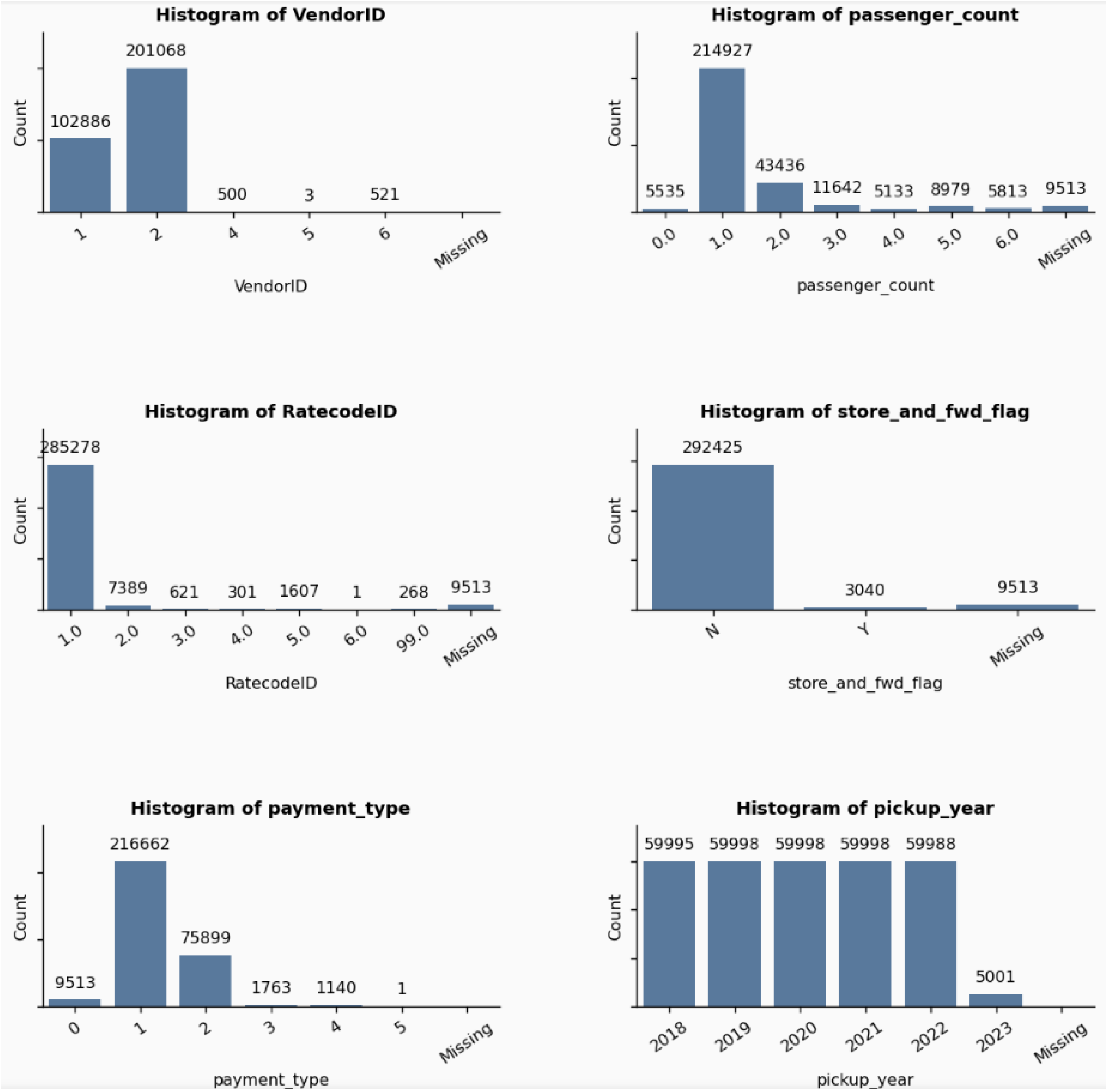
tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	RatecodeID	store_and_fwd_flag	fare_amount	extra	mta_tax
2020-04-13 11:19:51	2020-04-13 11:27:53	NaN	1.60	NaN	NaN	8.0	0.0	0.5
2020-04-13 11:19:51	2020-04-13 11:27:53	1.0	1.60	1.0	N	8.0	2.5	0.5
2020-04-20 07:55:55	2020-04-20 08:02:32	0.0	2.00	1.0	N	8.0	2.5	0.5
2020-04-20 07:55:55	2020-04-20 08:02:32	NaN	2.00	NaN	NaN	8.0	0.0	0.5
2021-01-06 13:32:53	2021-01-06 13:44:04	1.0	1.82	1.0	N	-9.0	0.0	-0.5
2021-01-06 13:32:53	2021-01-06 13:44:04	1.0	1.82	1.0	N	9.0	0.0	0.5

2. DATA QUALITY

2.3 Validity

Categorical columns have some abnormal values:

- **VendorID**: “4” and “5” are treated as “Others”
- **Passenger_count**: 0 and NA
- **RatecodeID**: NA
- **Store_and_fwd_flag**: NA
- **Payment_type:** confirmed that all NULLs of other columns happened only for Flex Fare trips (payment_type = 0)



2. DATA QUALITY

2.3 Validity

Numerical columns:

	count	mean	min	25%	50%	75%	max	std
tpep_pickup_datetime	304978	2020-07-16 11:54:50.074785792	2018-01-01 00:25:49	2019-04-08 09:18:22.500000	2020-07-17 17:01:39	2021-10-24 11:29:50.500000	2023-01-31 23:57:28	NaN
tpep_dropoff_datetime	304978	2020-07-16 12:11:31.905307648	2018-01-01 00:38:59	2019-04-08 09:32:34.750000128	2020-07-17 17:17:55	2021-10-24 11:42:53.249999872	2023-02-01 00:09:12	NaN
trip_distance	304978.0	4.587589	-16.86	1.0	1.73	3.21	177247.4	434.226624
fare_amount	304978.0	13.510189	-197.0	6.5	9.5	15.0	455.0	12.636651
extra	304978.0	0.903506	-6.0	0.0	0.5	1.0	18.5	1.185321
mta_tax	304978.0	0.492755	-0.5	0.5	0.5	0.5	2.54	0.074023
tip_amount	304978.0	2.224064	-20.0	0.0	1.86	2.95	115.56	2.7652
tolls_amount	304978.0	0.390715	-13.75	0.0	0.0	0.0	96.55	1.699547
improvement_surcharge	304978.0	0.312725	-1.0	0.3	0.3	0.3	1.0	0.110404
total_amount	304978.0	19.050501	-198.55	10.8	14.3	20.76	561.49	15.464014
congestion_surcharge	232346.0	2.24165	-2.5	2.5	2.5	2.5	2.75	0.791877
airport_fee	106217.0	0.085815	-1.25	0.0	0.0	0.0	1.25	0.318628

Negative values are present in small numbers in numeric columns (less than 0.4%).

```
Negative value count and % per column:
trip_distance: 9 (0.00%)
fare_amount: 1,162 (0.38%)
extra: 541 (0.18%)
mta_tax: 1,123 (0.37%)
tip_amount: 14 (0.00%)
tolls_amount: 38 (0.01%)
improvement_surcharge: 1,160 (0.38%)
total_amount: 1,167 (0.38%)
congestion_surcharge: 891 (0.29%)
airport_fee: 55 (0.02%)
```

2. DATA QUALITY

2.3 Validity

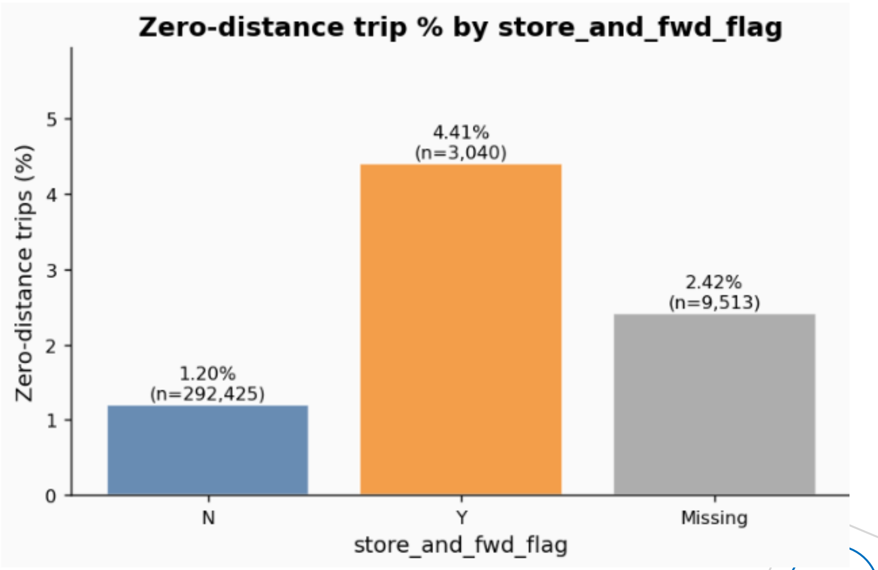
Numerical columns:

Values of 0 in numerical columns:

```
Zero value count and % per column:
trip_distance: 3,882 (1.27%)
fare_amount: 129 (0.04%)
extra: 134,912 (44.24%)
mta_tax: 2,196 (0.72%)
tip_amount: 91,871 (30.12%)
tolls_amount: 287,172 (94.16%)
improvement_surcharge: 151 (0.05%)
total_amount: 78 (0.03%)
congestion_surcharge: 22,228 (7.29%)
airport_fee: 98,815 (32.40%)
```

- Some possible explanations for the values of 0:
  - **trip_distance**: can be 0 for cancelled trips, GPS error, or very short trips (rounded to 0), etc.
  - **fare_amount** and **total_amount**: can be 0 for free trips (eg. cancelled trip or technical error or other reasons).
  - Other columns can have valid 0 values.

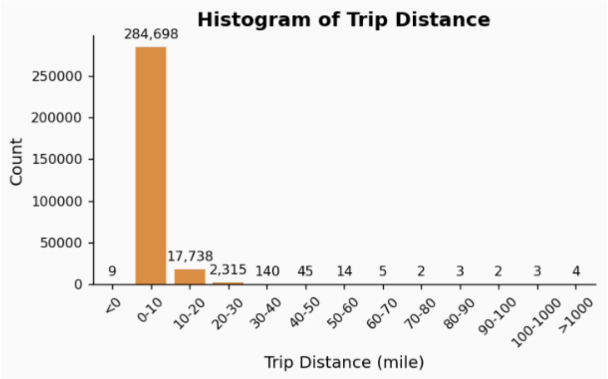
- **Store-and-forward** trips exhibit a substantially higher rate of **zero-distance** records (4.41%) compared with non-store-and-forward trips (1.20%)



2. DATA QUALITY

2.4 Accuracy

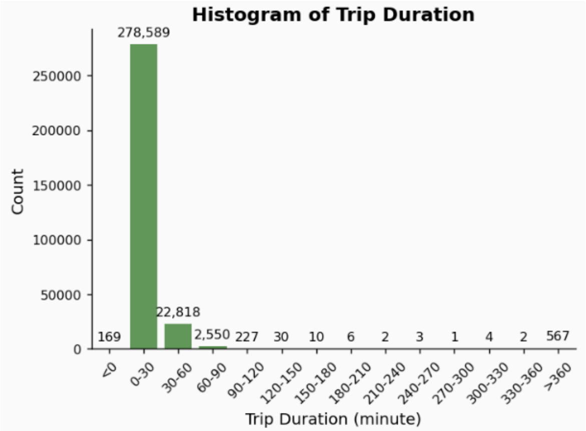
- Implausible trips found: negatives, >100 miles, >3 hours duration, >100 mph speed

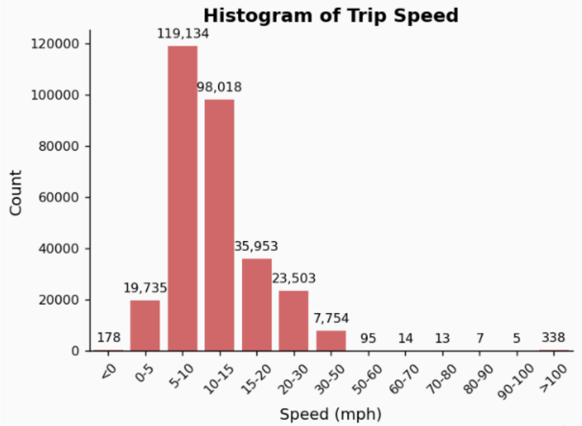


- **total_amount** mismatches (vs sum of components) are linked to **congestion_surcharge** recording errors (confirmed by chi-square test)

```
Distribution of congestion_surcharge when mismatch is -2.50:
congestion_surcharge
 2.5    66797
 0.0        3
-2.5        3

Distribution of congestion_surcharge when mismatch is +2.50:
congestion_surcharge
NaN    4292
2.5      94
0.0       3
```





3. DATA CLEANING

3.1 Completeness: NULL imputation

- passenger_count, RatecodeID, store_and_fwd_flag: mode-imputed with a missingness flag column added
- congestion_surcharge and airport_fee: retained as NULL (not forced to 0)

3.2 Uniqueness: duplicate removal

- Deleted the NULL-metadata row from each of the 2 metadata duplicate pairs
- Deleted the negative-value row from the 1 financial duplicate pair

3.3 Validity: remove negative values

- Dropped rows with negative trip_distance and financial columns
- Zero values retained — removing them silently would undercount minimum-fare and cancellation revenue

3.4 Accuracy: plausibility filters

- Kept only trips with: distance ≤100 miles, duration ≤180 minutes, speed ≤100 mph
- total_amount mismatches not corrected - no reliable business rule to determine which field is wrong

Result

- 0.82% of rows removed; cleaned dataset retains over 302k trips with more reasonable distributions

4. FEATURE ENGINEERING

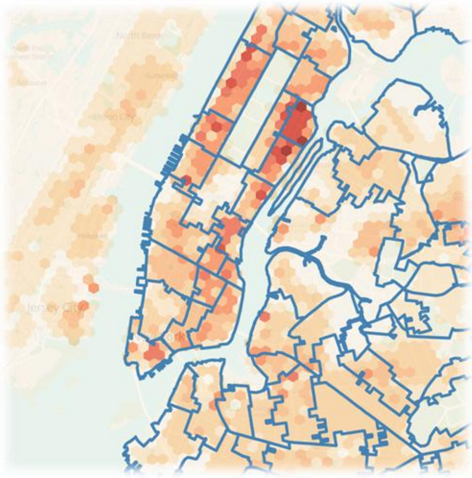
4.1 New datetime and derived columns

- Pickup year, month, day-of-week, hour of day - for temporal analysis
- trip_duration_min, avg_speed_mph - for efficiency analysis
- Human-readable labels for VendorID, RatecodeID, payment_type

4.2 Geospatial enrichment

- Joined TLC zone lookup table: added pickup/dropoff borough, zone name, and service zone
- Loaded TLC zone shapefiles for geographic heatmap visualizations

LocationID	Borough	Zone	service_zone
1	EWR	Newark Airport	EWR
2	Queens	Jamaica Bay	Boro Zone
3	Bronx	Allerton/Pelham Gardens	Boro Zone
4	Manhattan	Alphabet City	Yellow Zone
5	Staten Island	Arden Heights	Boro Zone



4. FEATURE ENGINEERING

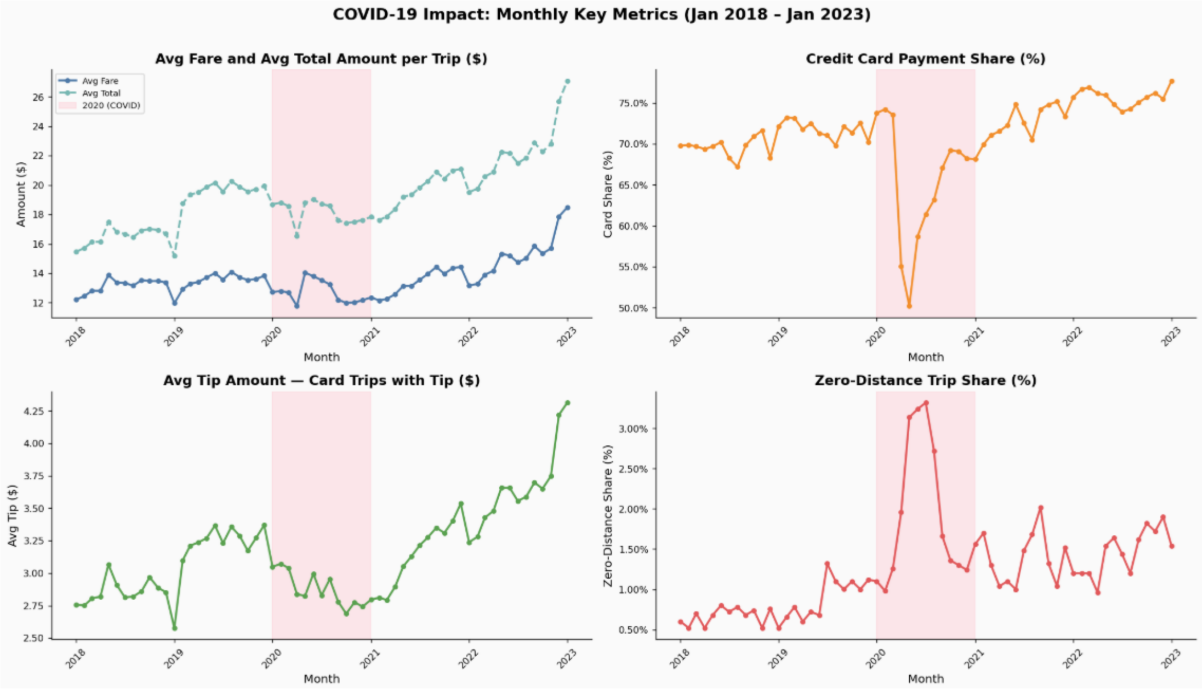
4.3 Sample Validation

Representativeness:

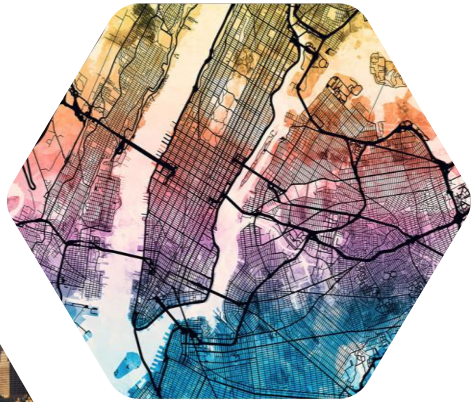
- Tested payment mix, borough share, and hourly distribution stability across months
- Sample composition is internally consistent; ratio metrics are suitable for trend analysis

COVID Impact:

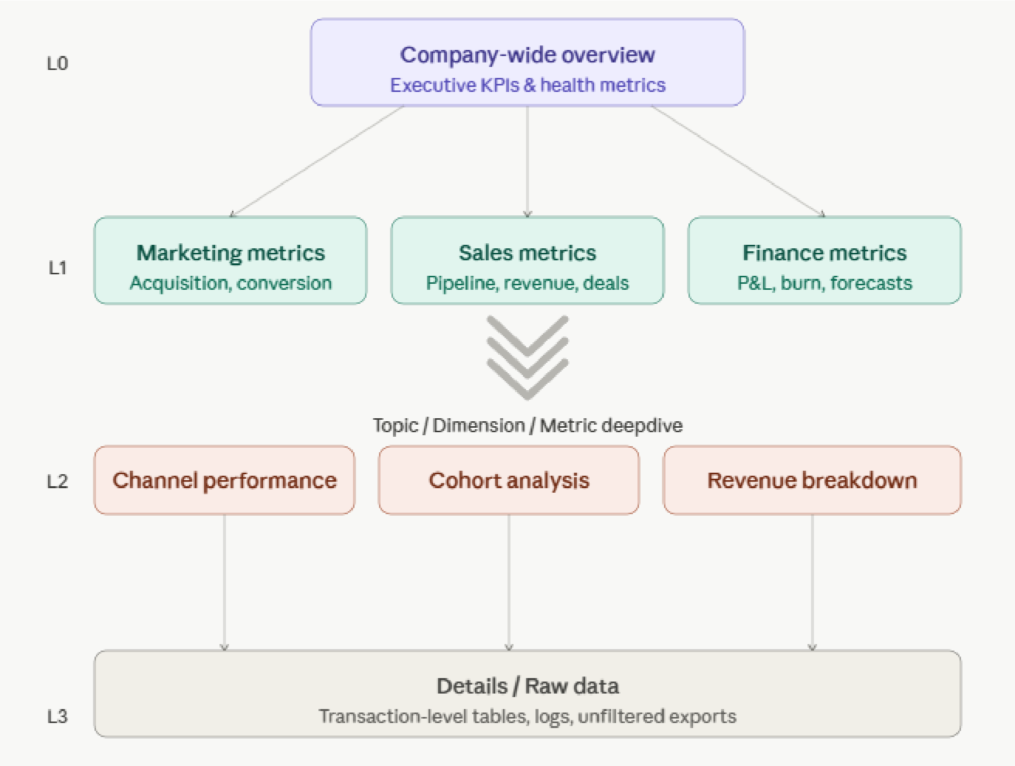
- Many metrics experienced disruption in 2020
- All Correlation with COVID-19 noted; causality cannot be established from this sample alone

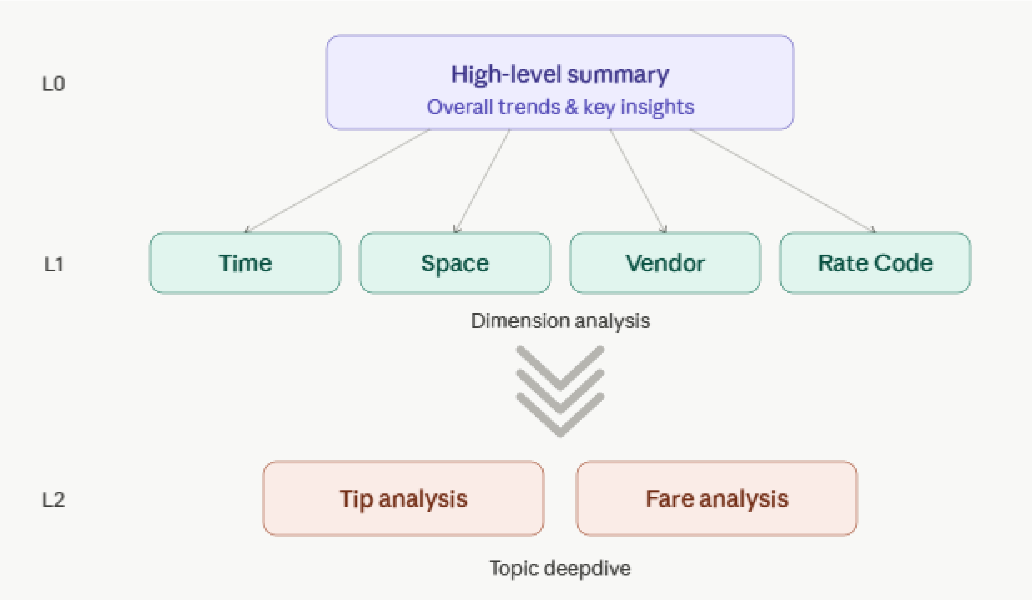


OUTLINE



5. DATA ANALYSIS





On the left is my general data analysis framework. In this assignment, I used the modified framework (on the right).

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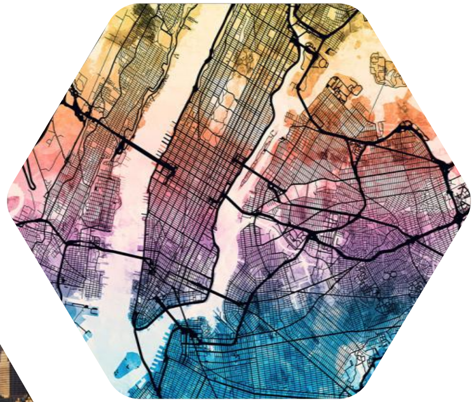
    - 5.1 L0 High-level overview
    - 5.2 L1 Time
    - 5.3 L1 Space
    - 5.4 L1 Vendor
    - 5.5 L1 Rate code
    - 5.6 L2 Tip
    - 5.7 L2 Fare
6. Conclusions

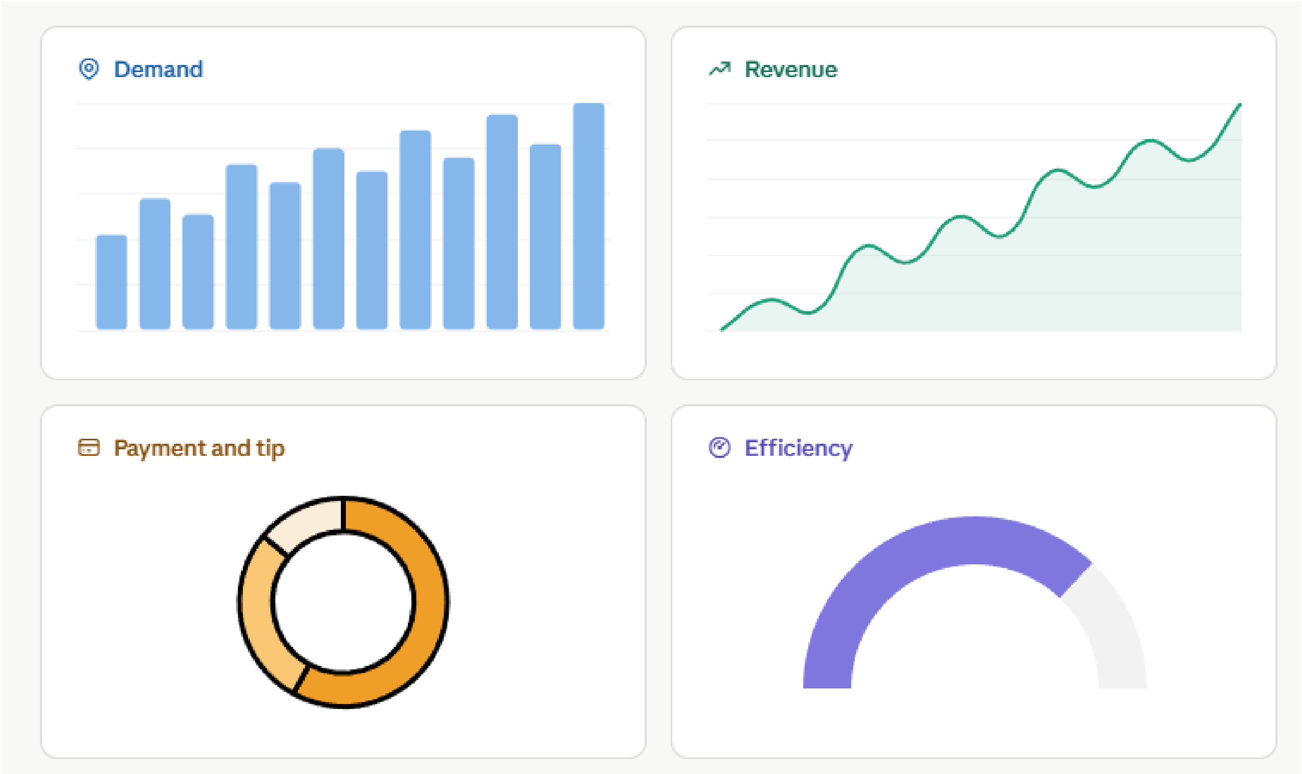


5. DATA ANALYSIS

Map of New York City with 5 boroughs and 3 airports (JFK, La Guardia & Newark)

5. DATA ANALYSIS

5.1 L0 High-level Overview



The metrics are presented in 4 main groups as above.

5. DATA ANALYSIS

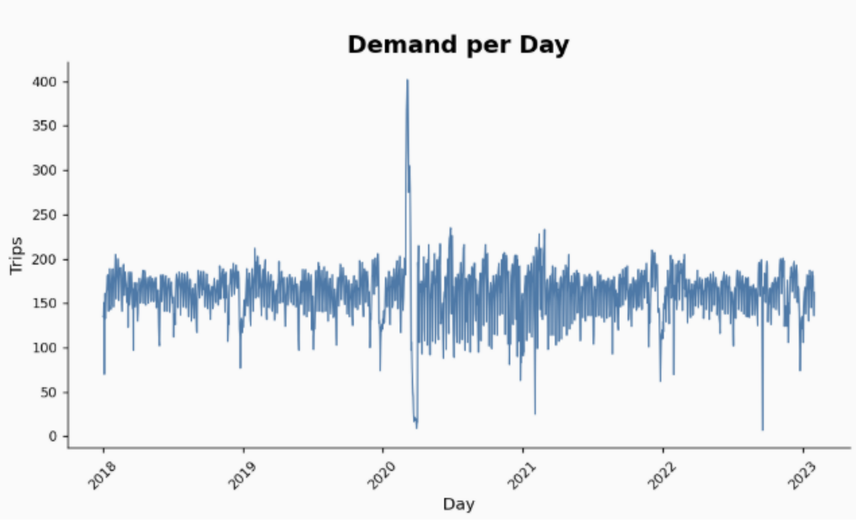
5.1 L0 High-level Overview

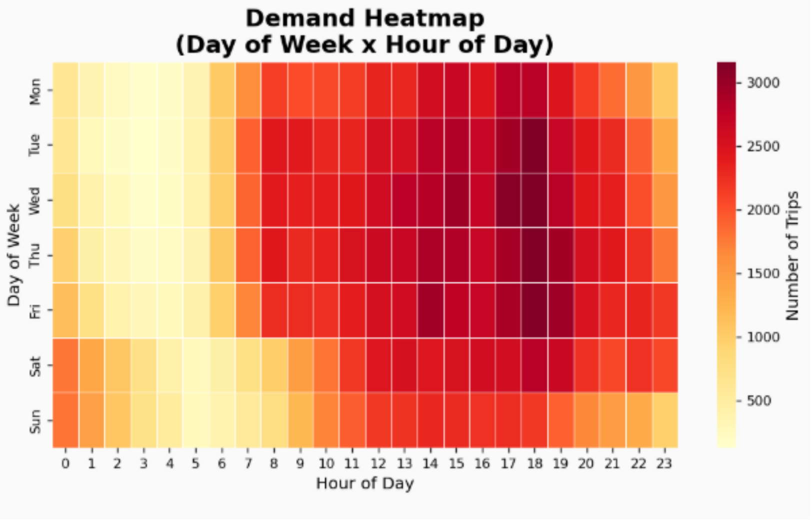
a) Demand

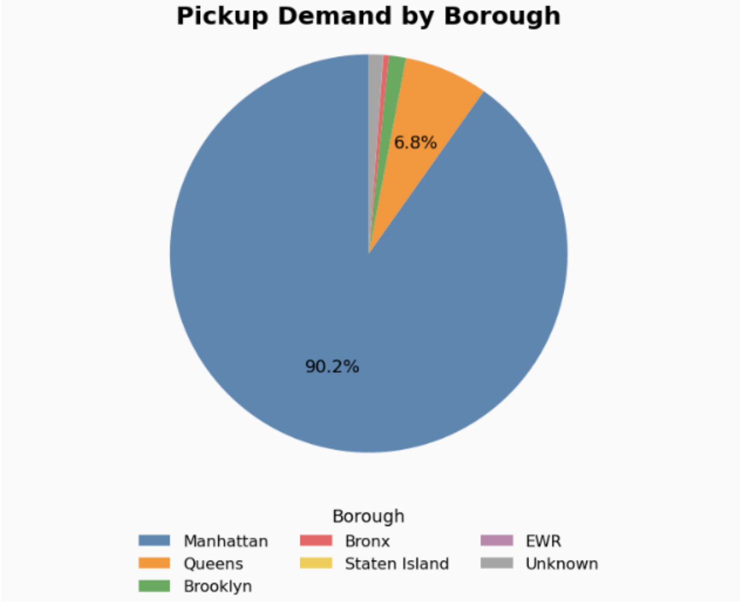
There is disruption in demand around March 2020 (will be discussed in 5.2).

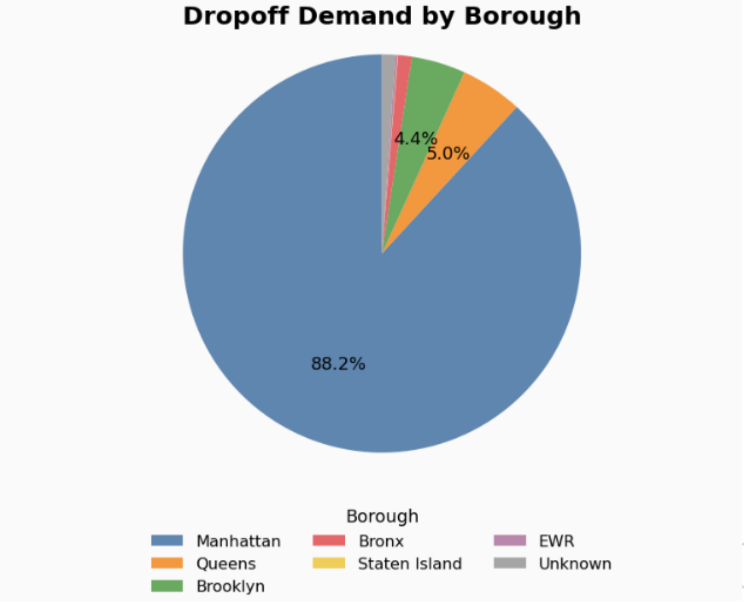
The hourly profile indicates clear late-afternoon and early-evening intensity, while day-of-week results show stronger demand on weekdays than weekends.

Demand is the highest in Manhattan (about 90% of total number of trips).







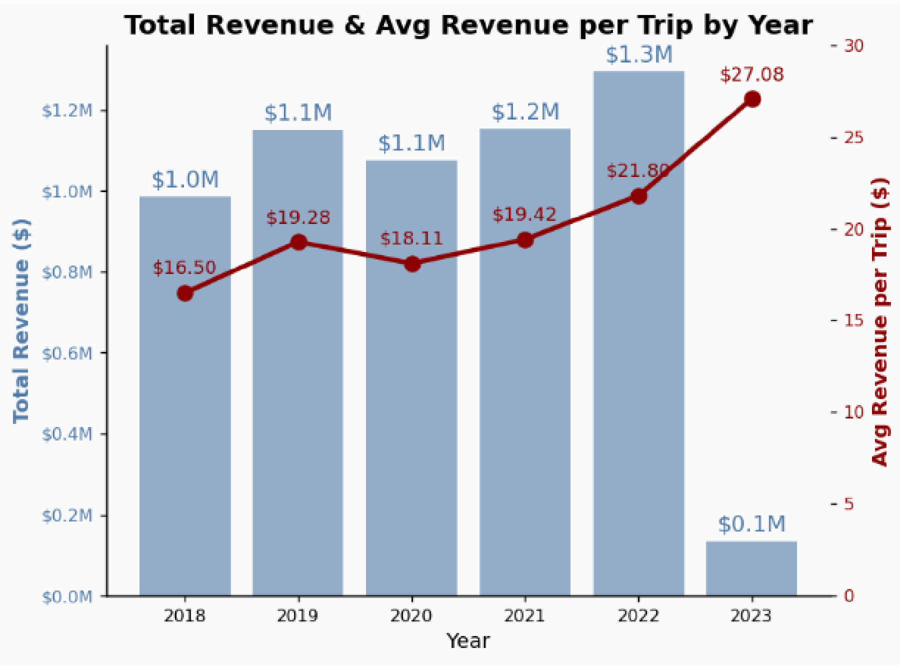


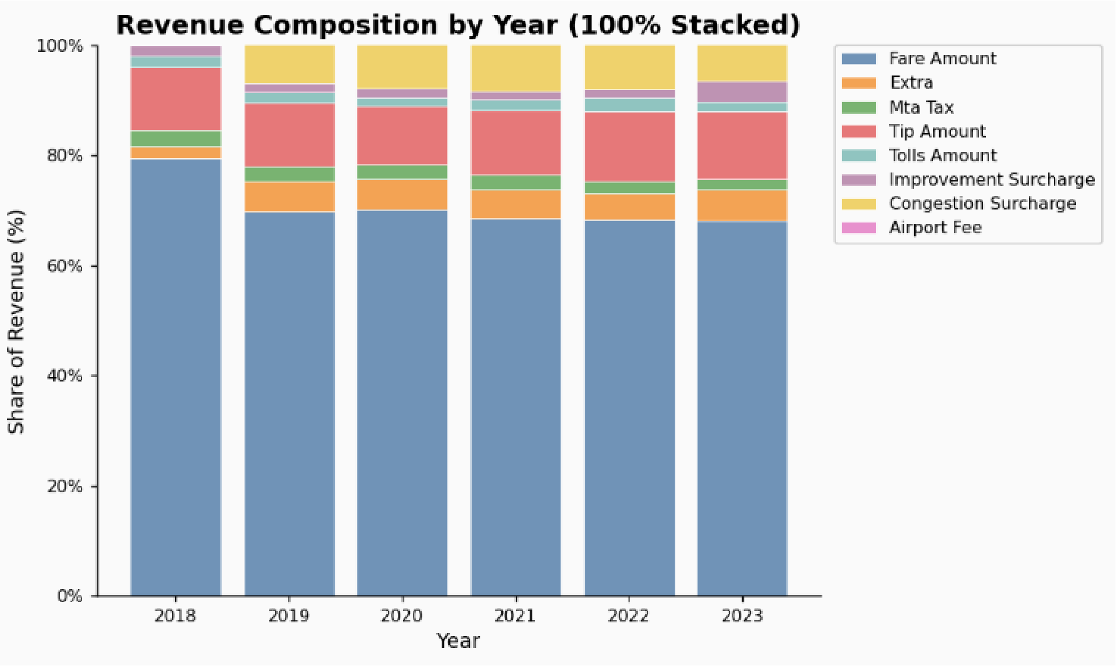
5. DATA ANALYSIS

5.1 L0 High-level Overview

b) Revenue

- Average revenue per trip trends upward and peaks in 2023.
- Revenue composition remains fare-led, with tip amount as the second-largest contributor.
- Congestion surcharge seems to be introduced in 2019.



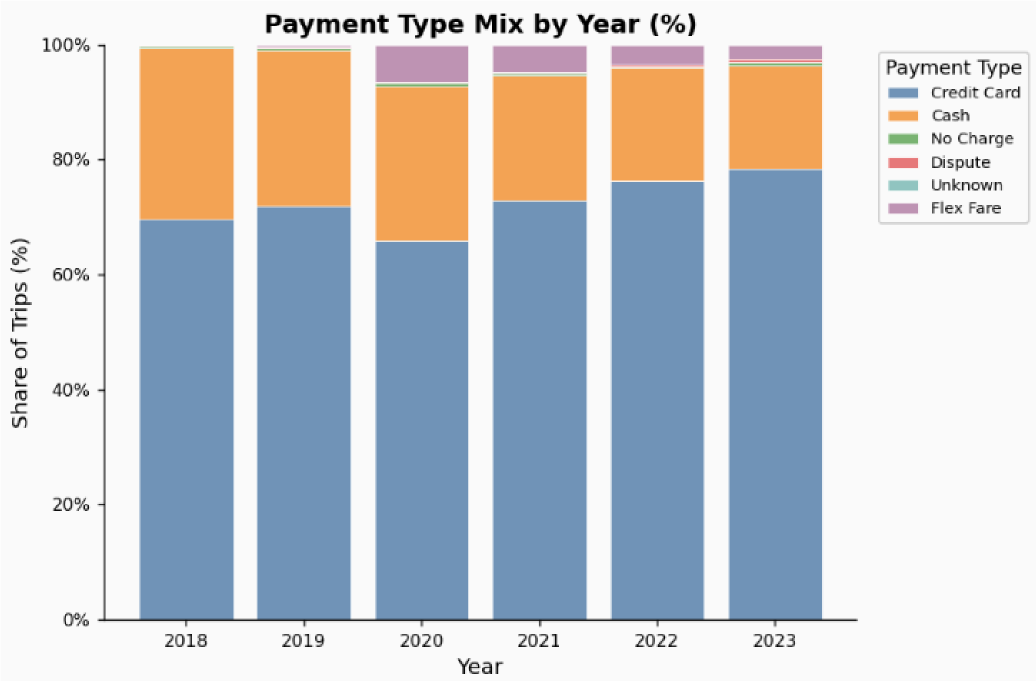


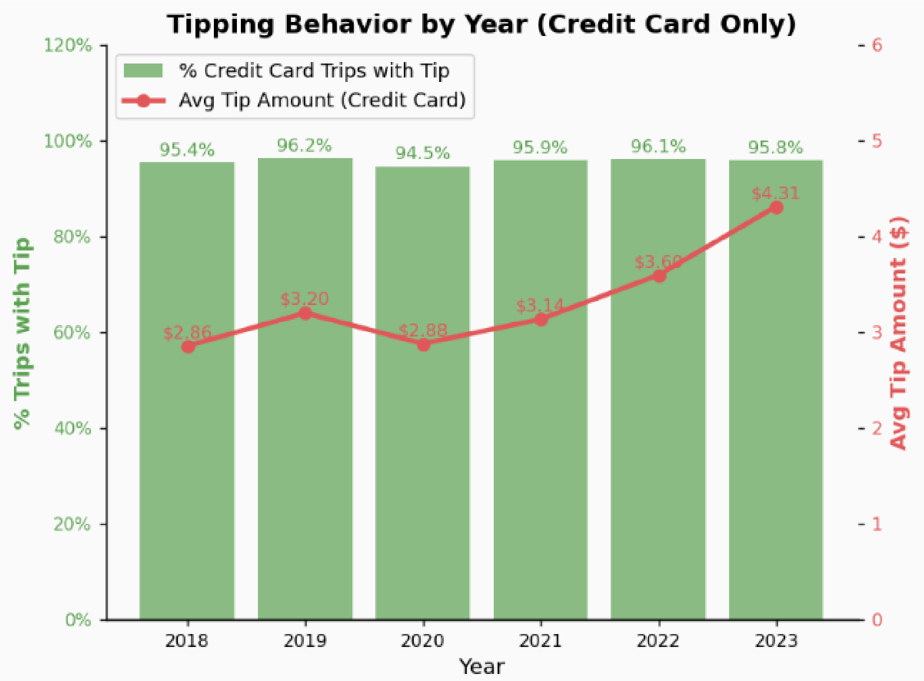
5. DATA ANALYSIS

5.1 L0 High-level Overview

c) Payment & Tip

- Card payments dominate across all years and increase over time, while cash share generally declines.
- The share of tipped trips remained stable at around 95%.
- Average tip amount also trends upward (roughly $2.9 to $4.3), with a dip in 2020 and strong growth afterward.
- Overall, payment behavior is becoming more cash-less and tip amount is improving over time.





5. DATA ANALYSIS

5.1 L0 High-level Overview

d) Efficiency

- Trip duration slightly dipped in 2020 even though distance did not change much in that year.

- The heatmap shows clear time-dependency, with faster speeds after midnight and slower speeds during office hours on weekdays.

- Revenue per mile and per minute jumped significantly in 2023.

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    5.1 L0 High-level overview

    5.2 L1 Time

    5.3 L1 Space

    5.4 L1 Vendor

    5.5 L1 Rate code

    5.6 L2 Tip

    5.7 L2 Fare

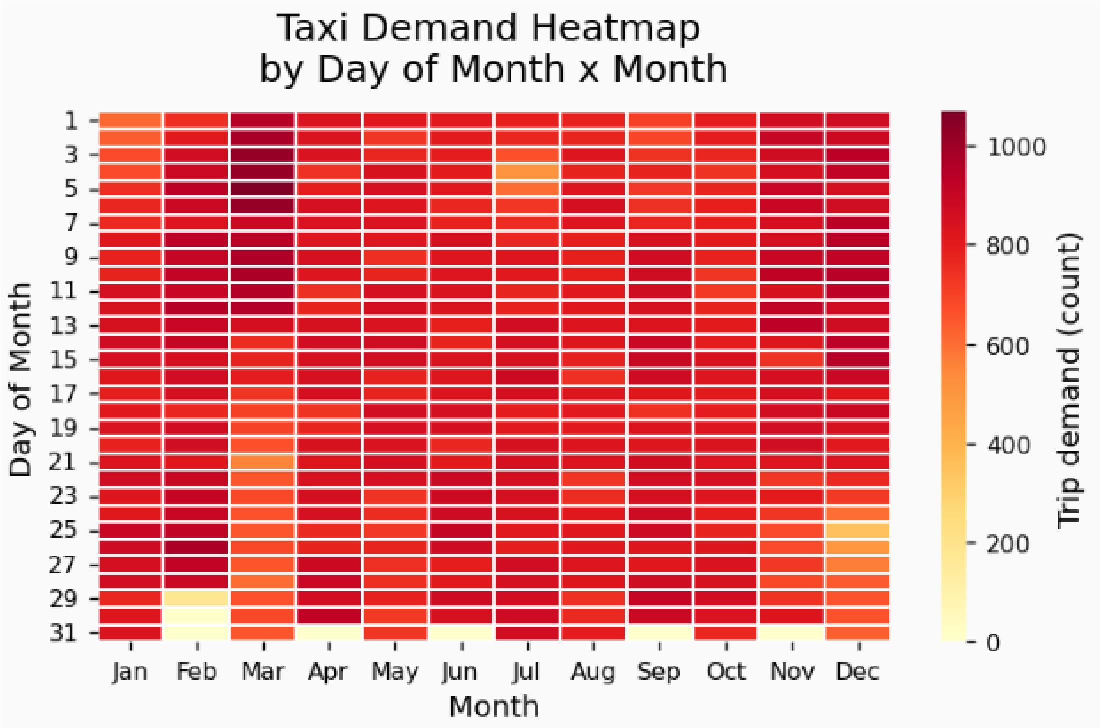
6. Conclusions

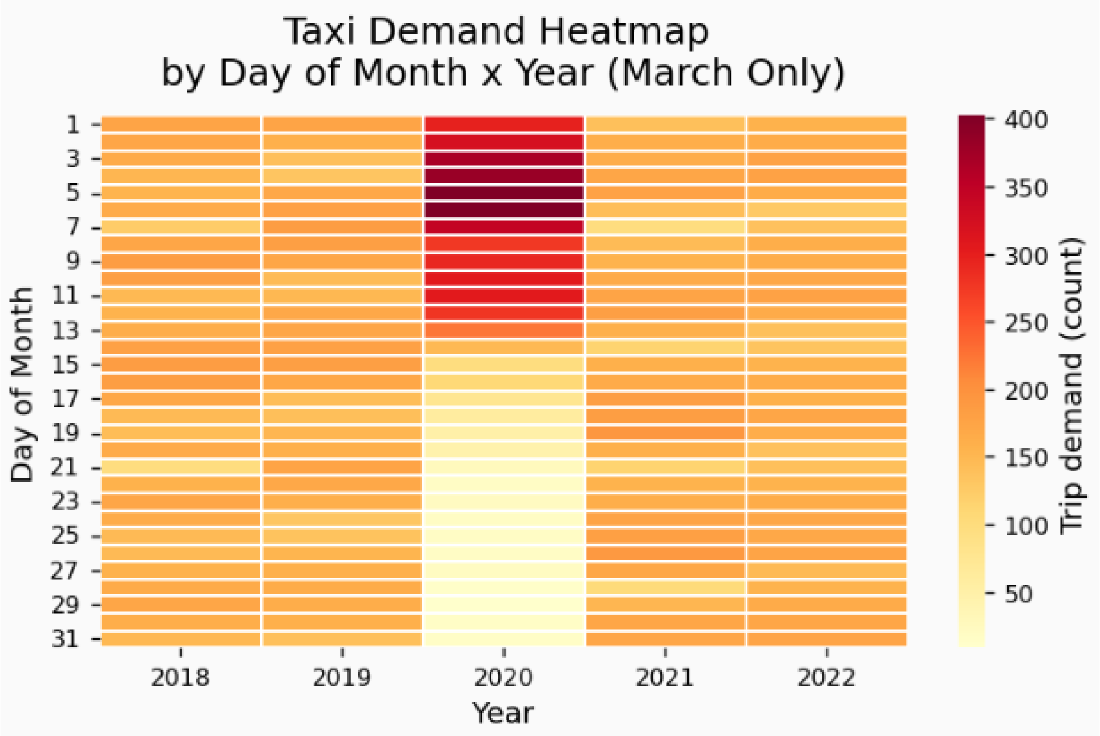
5. DATA ANALYSIS

5.2 L1 Time

- The heatmap on the left show that there is lower demand on the holidays (4 July and Christmas week). March seems highly concentrated in the early days of the month.

- The heatmap on the right explains that 2020 is the outlier, which correlates with Covid timing when there was no demand in late March 2020.





5. DATA ANALYSIS

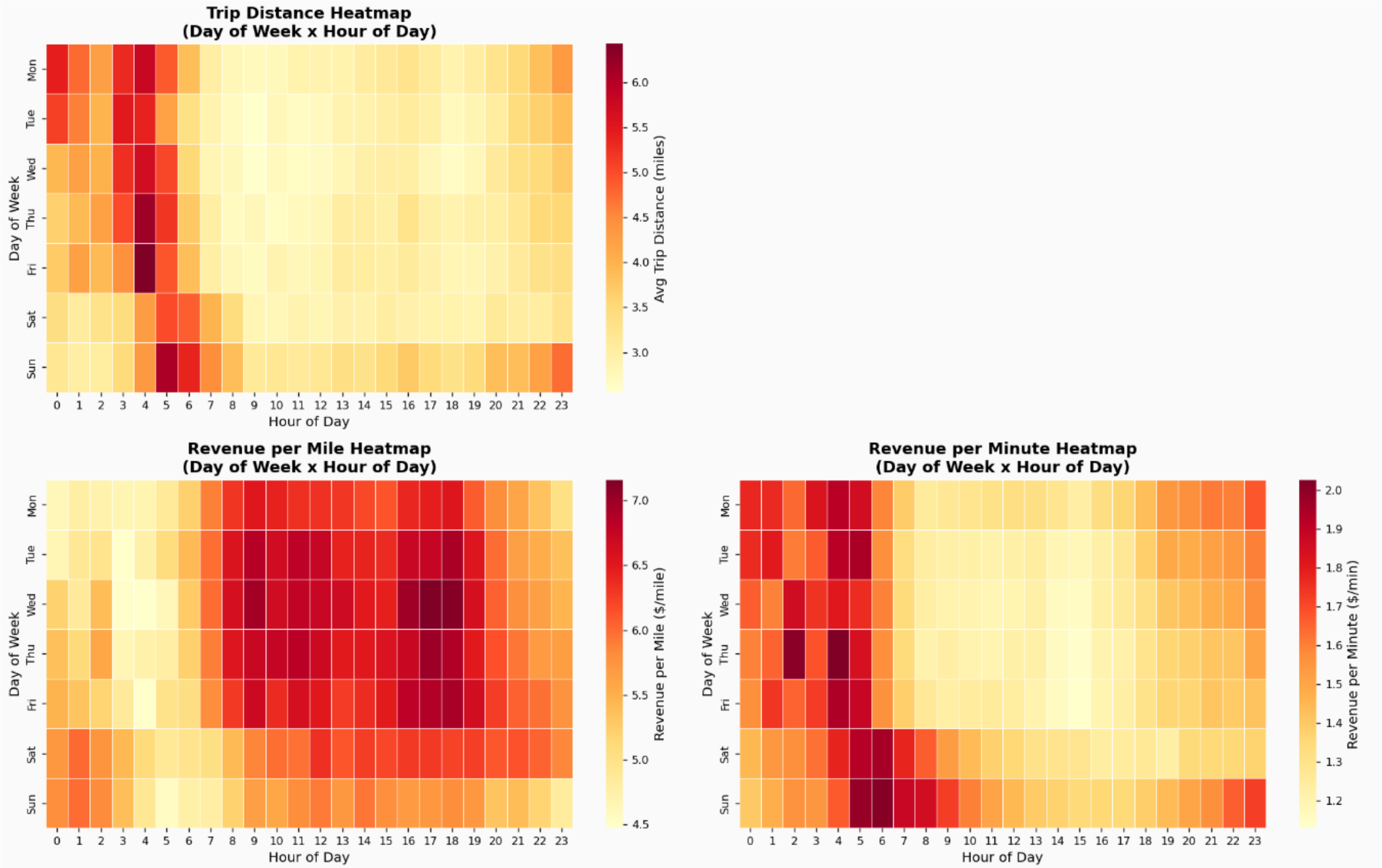
5.2 L1 Time

- Trip distance is highest in late-night to early-morning windows, especially around 3-6 AM.

- In contrast, Revenue per mile is weaker in overnight periods.

- Revenue per minute has opposite pattern to Revenue per mile.

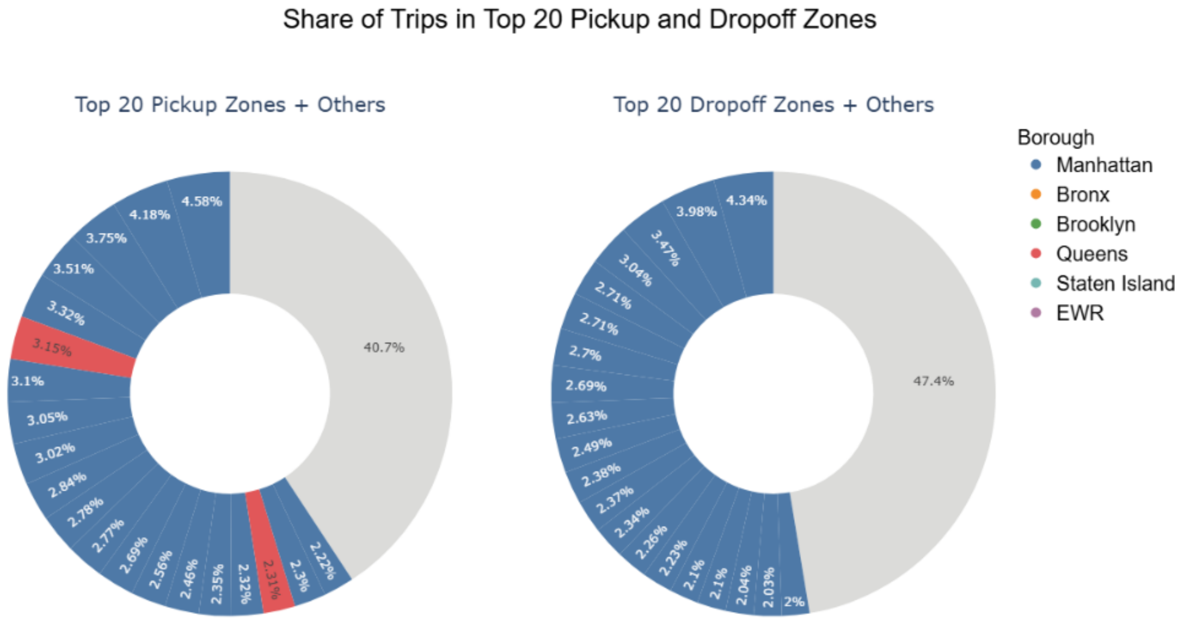
- Together, the three heatmaps suggest distance influence revenue efficiency. Overnight periods have longer-distance trips and better Revenue per minute efficiency, but lower Revenue per mile.

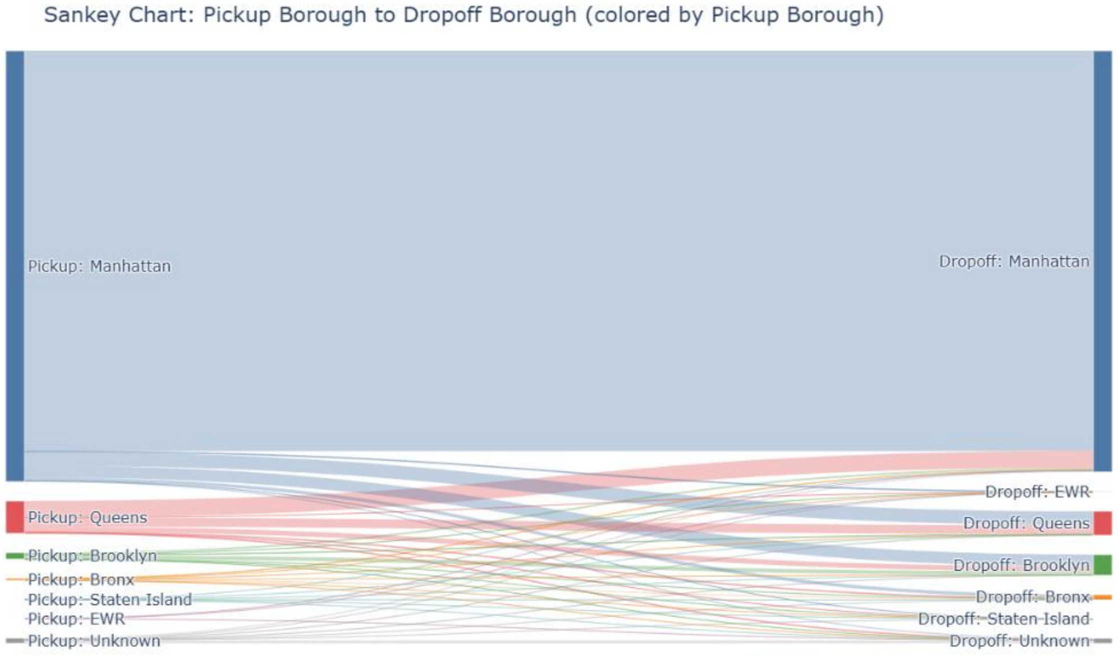


5. DATA ANALYSIS

5.3 L1 Space

- The trips are **highly concentrated** in a few zones: out of ~250 zones, both the top 20 pickup and dropoff zones account for more than half of trip volume.
- The top 20 pickup zones include 18 in Manhattan and 2 airports in Queens.
- The top 20 dropoff zones are all in Manhattan.
- Nearly 85% of trips start and end within Manhattan, which is expected given that Manhattan is the most densely populated borough and a major commercial hub.





5. DATA ANALYSIS

5.3 L1 Space

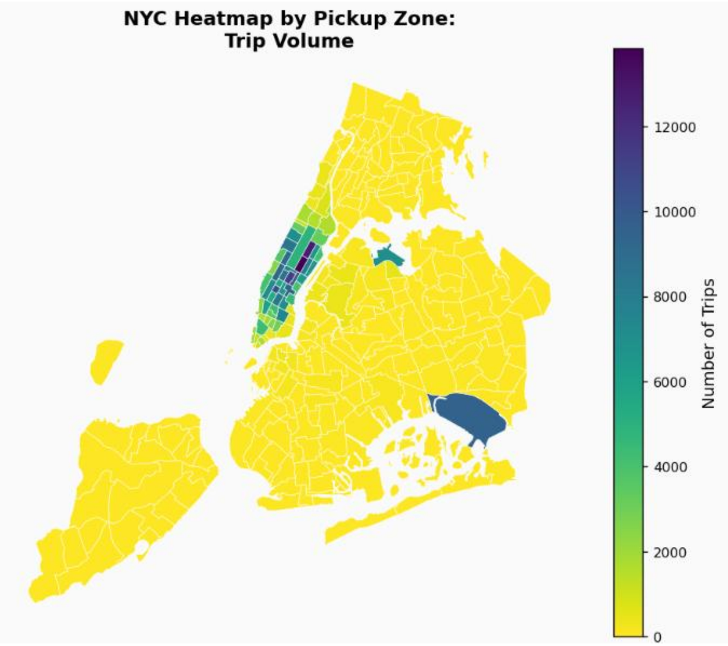
- **Trip volume** is heavily concentrated in Manhattan, with airport-related zones in Queens as secondary hotspots.

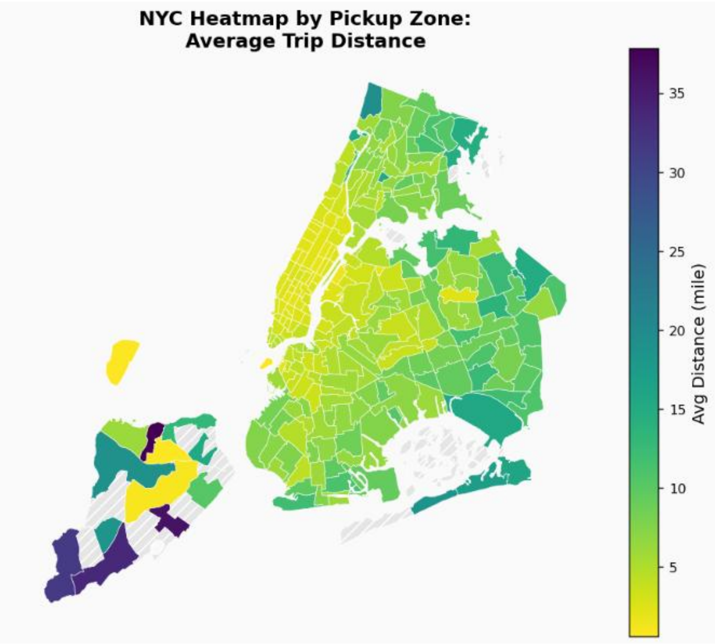
- **Average trip distance** is generally shorter in high-volume core zones (especially Manhattan) and longer in lower-volume outer-borough zones.

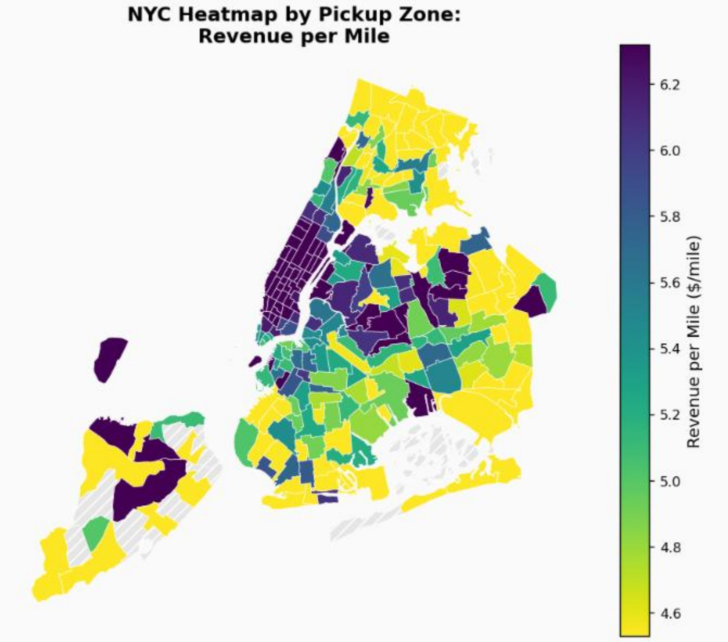
- **Revenue per mile** tends to be strongest in central, dense activity zones and weaker across many peripheral zones.

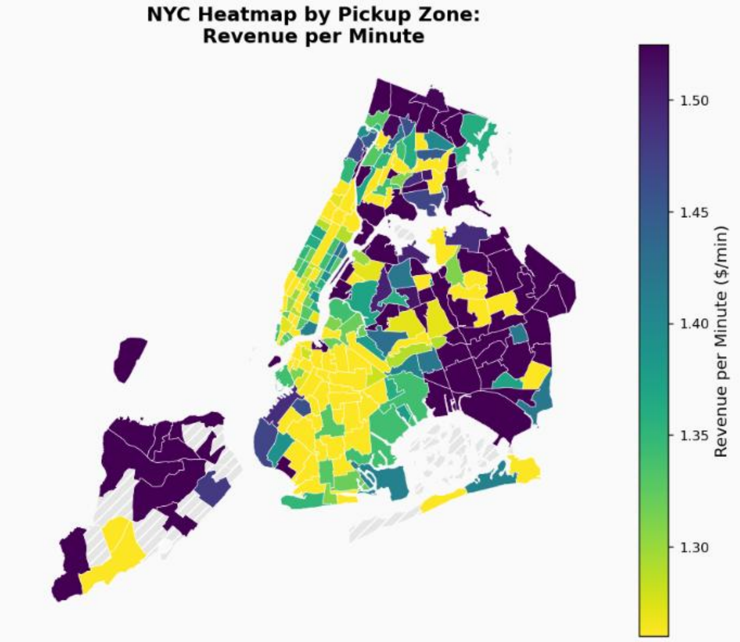
- **Revenue per minute** adds a complementary view: the general pattern is opposite to Revenue per mile.

- Together, the four maps show a clear core-periphery pattern: a high-demand urban core with stronger distance-based yield versus a lower-demand outer area with more mixed efficiency outcomes.





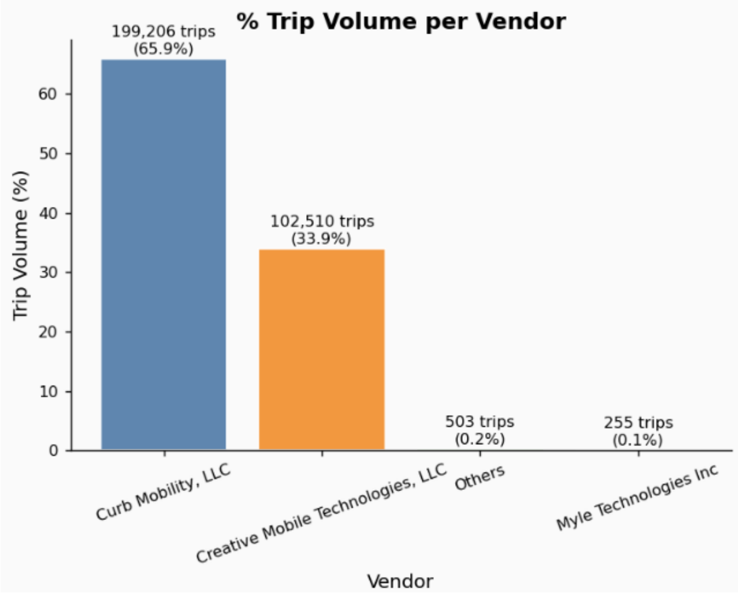


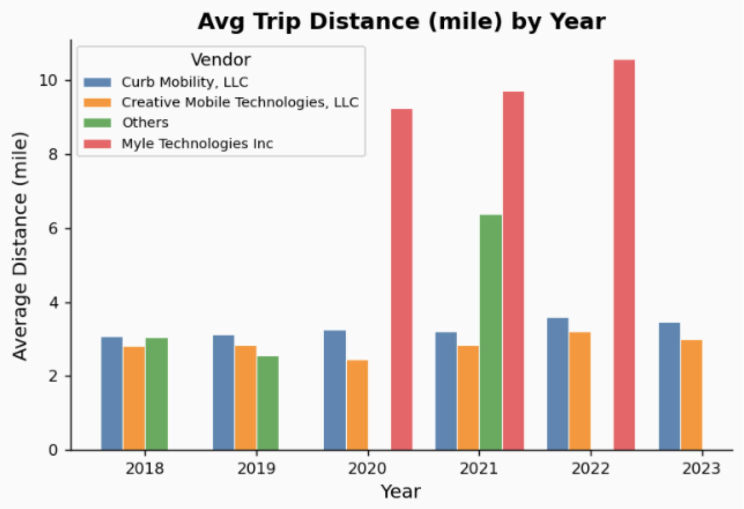


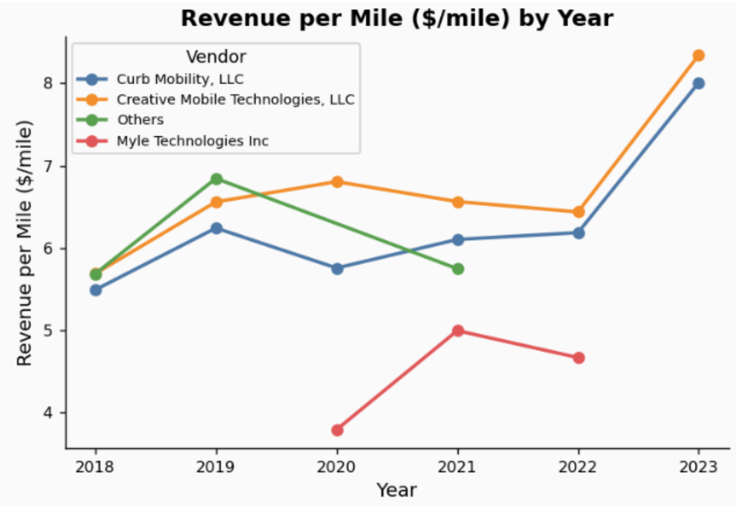
5. DATA ANALYSIS

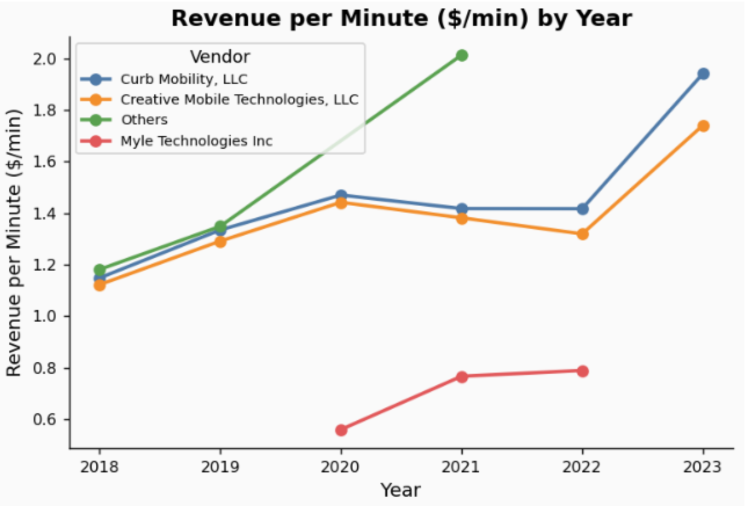
5.4 L1 Vendor

- Trip volume is highly concentrated in Curb Mobility and Creative Mobile Technologies; other vendors have minimal share.
- The two major vendors maintain stable trip distance, while smaller vendors tend to run longer trips.
- Revenue per mile and per minute is stronger for the top vendors and jumps in 2023.
- Myle remains lower for both efficiency metrics, suggesting a different fare mix.
- Results for small vendors are volatile and should be interpreted cautiously due to low trip counts.









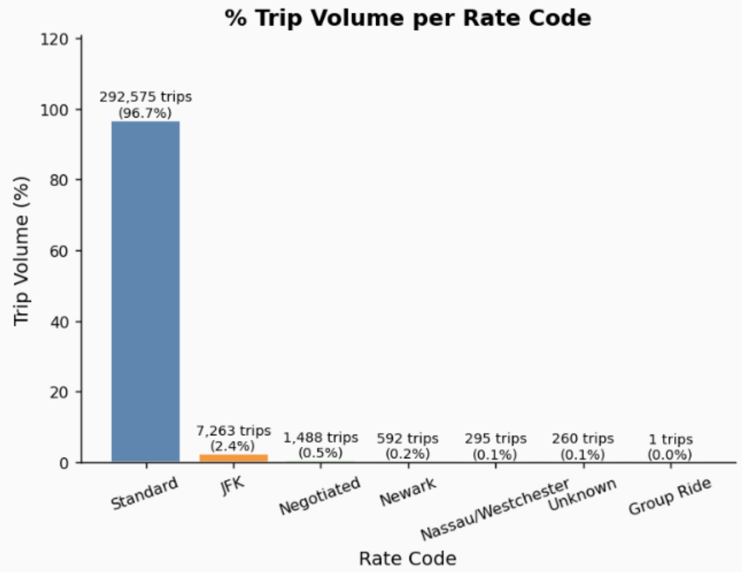
5. DATA ANALYSIS

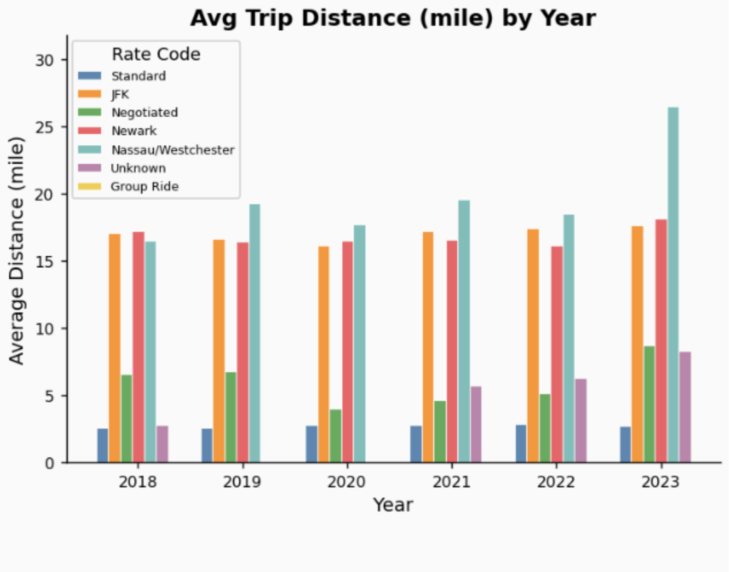
5.5 L1 Rate Code

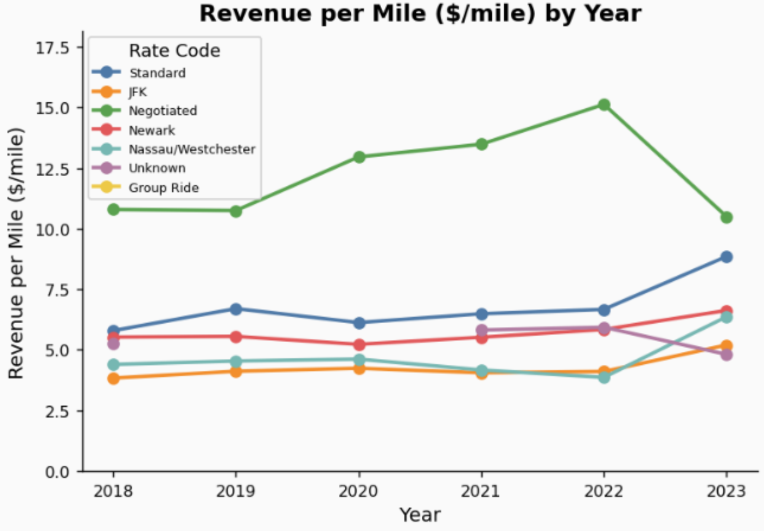
- Standard rides drive the business by volume, while non-standard codes remain niche.

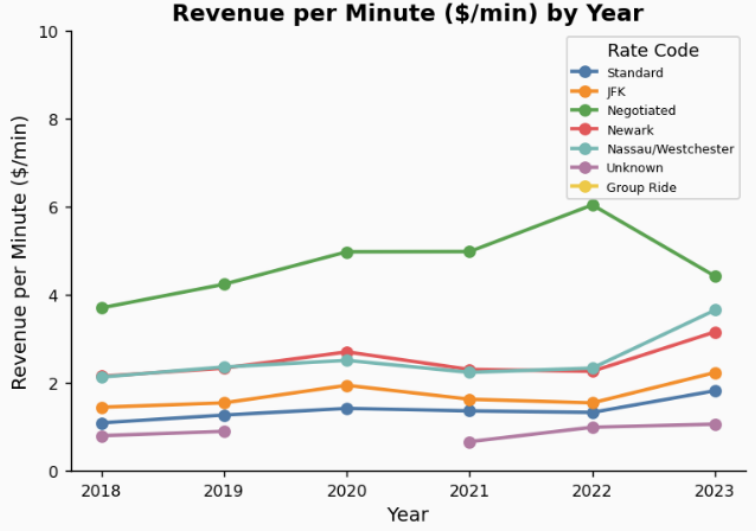
- Airport and codes carry longer trip profiles.

- Negotiated rate code perform the best for both revenue efficiency metrics.



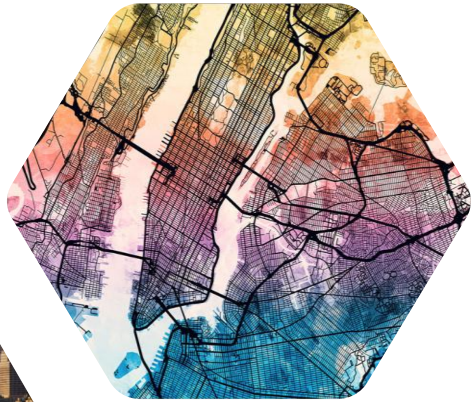






OUTLINE

1. Introduction
2. Data quality
3. Data cleaning
4. Feature engineering
5. Data analysis
    - 5.1 L0 High-level overview
    - 5.2 L1 Time
    - 5.3 L1 Space
    - 5.4 L1 Vendor
    - 5.5 L1 Rate code
    - 5.6 L2 Tip
    - 5.7 L2 Fare
6. Conclusions



5. DATA ANALYSIS

5.6 L2 Tip

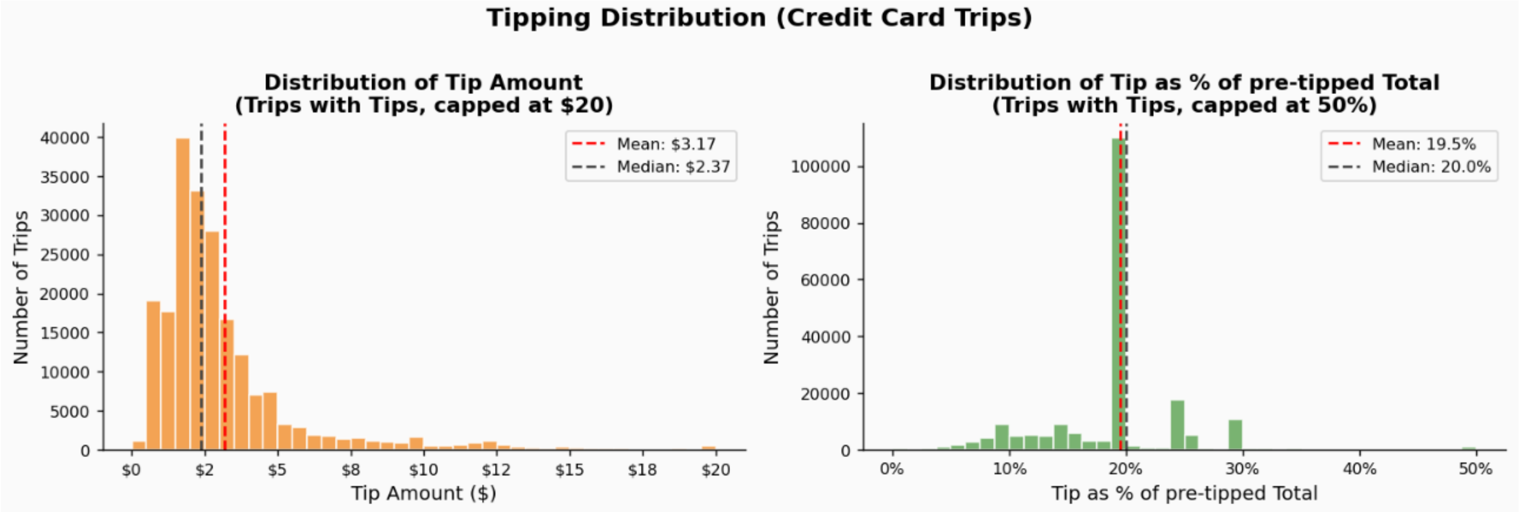
- Tips are right-skewed: most trips have low-to-moderate tips, while high tips are relatively rare.

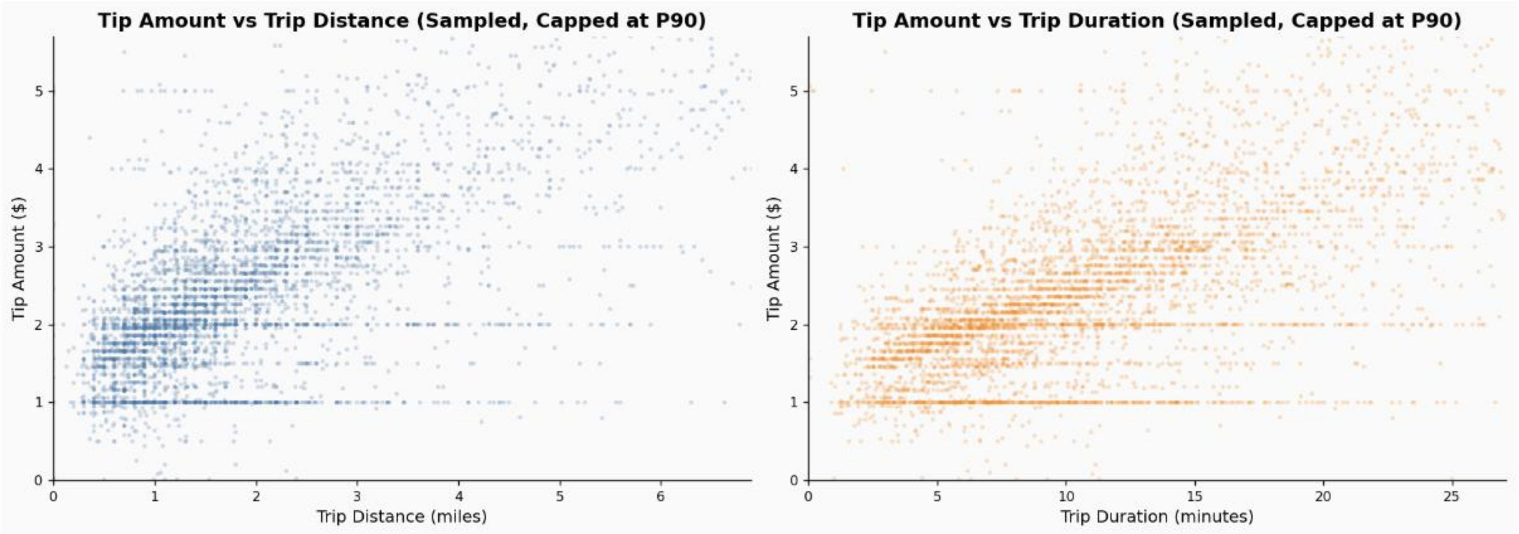
- The concentration near small tip values suggests many short trips or low-fare rides. A thin long tail indicates occasional high-tip trips, likely linked to longer distances.

- A large majority of trips have 20% tip. This may be the default choice for customers.

- Many trips have tip amount at $1 and $2.

- Positive correlation between tip amount vs trip distance and trip duration (Spearman coefficient ~0.68).

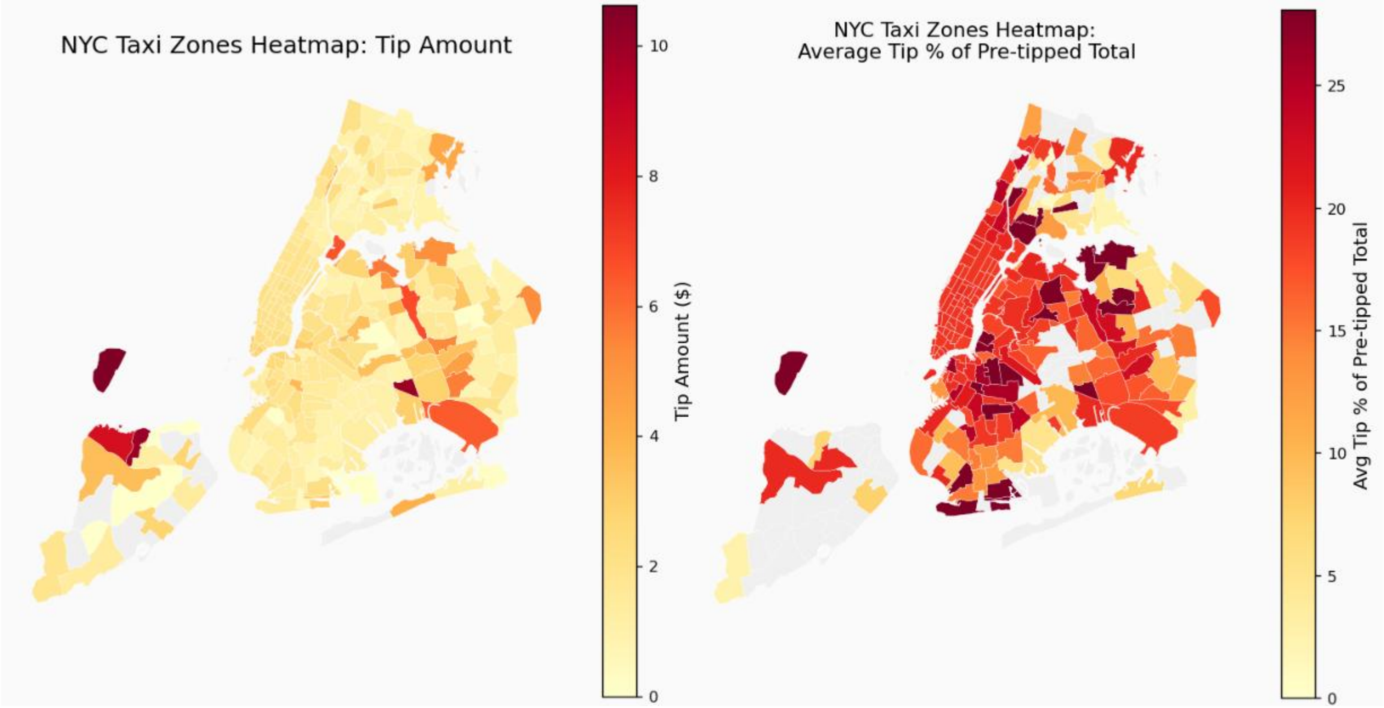


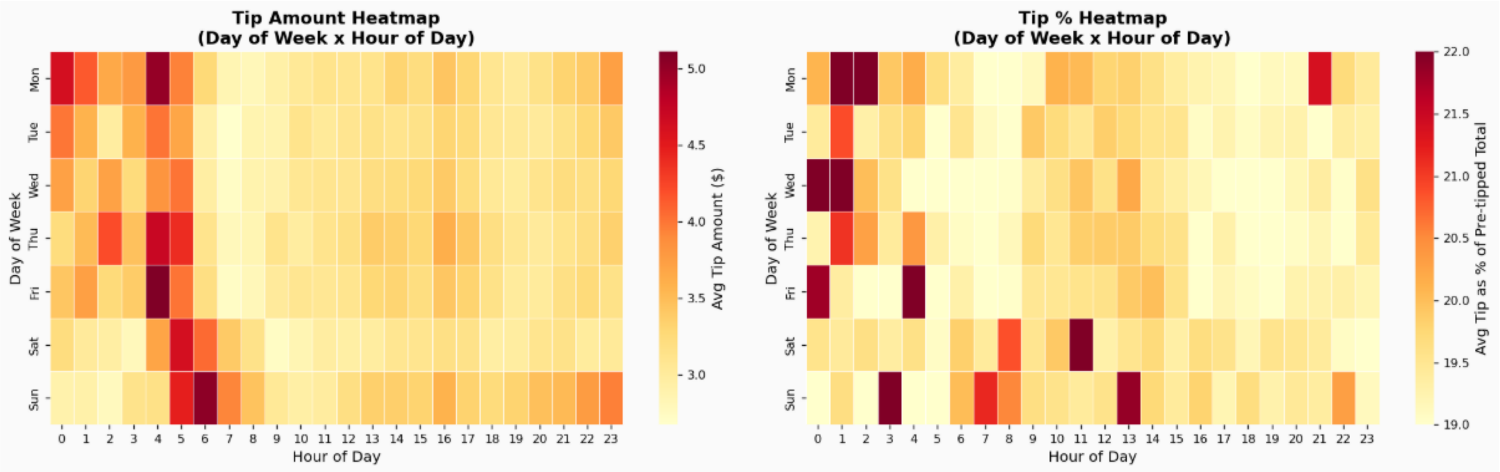


5. DATA ANALYSIS

5.6 L2 Tip

- Zones with the highest tips are mostly in peripheral areas and the airports.
- Tip amounts tend to be higher after midnights.

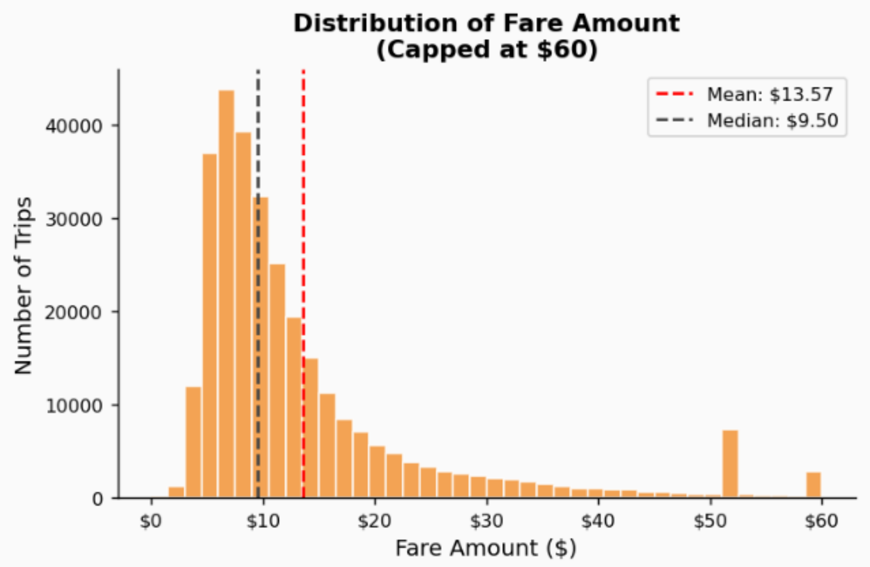


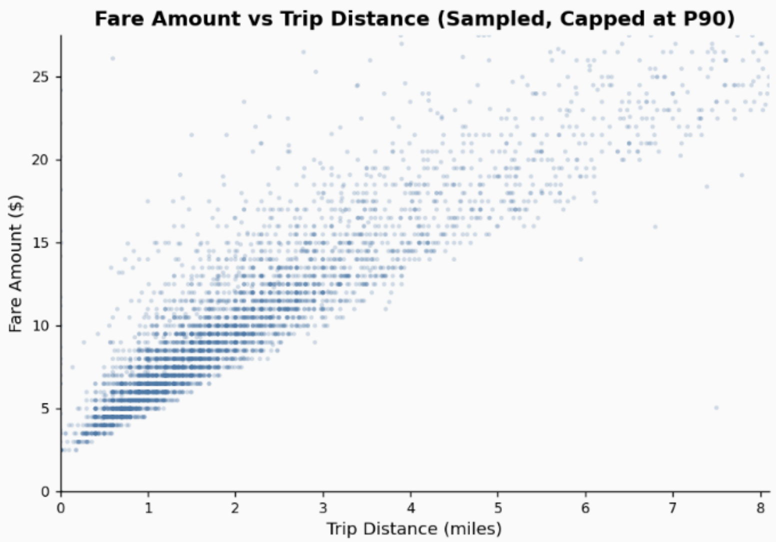


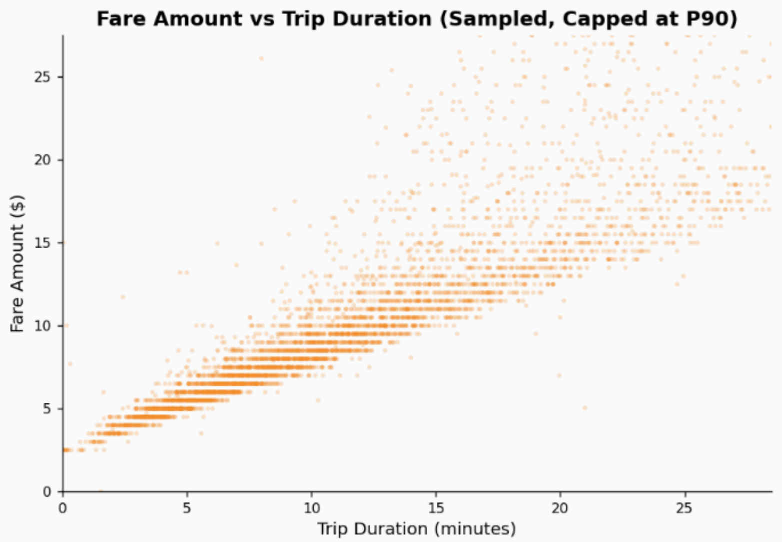
5. DATA ANALYSIS

5.7 L2 Fare

- The distribution of fare amounts is right-skewed, with a long tail of higher fares.
- Fare rises with both trip distance and trip duration, confirmed by strong positive relationships in both scatter plots and Spearman test (coefficient ~0.9).
- The fare-distance pattern is tighter and more linear than fare-duration, suggesting distance is the cleaner direct driver of pricing.
- The fare-duration plot is more dispersed at the top right, indicating that other factors introduce additional fare variability beyond travel time alone.







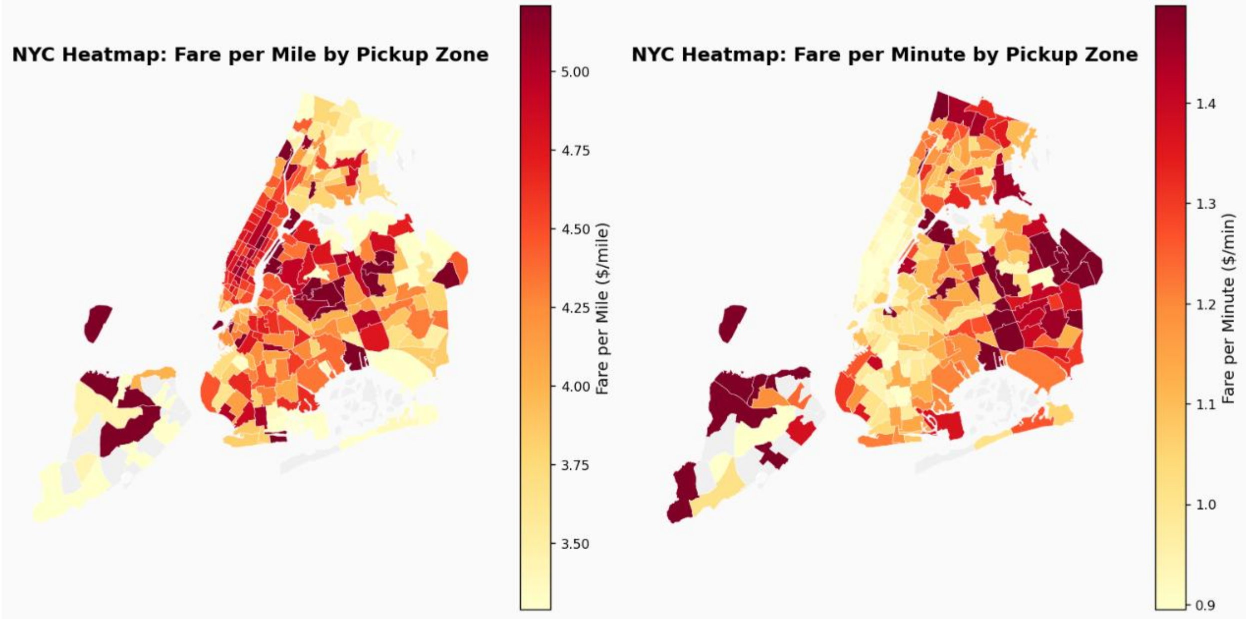
5. DATA ANALYSIS

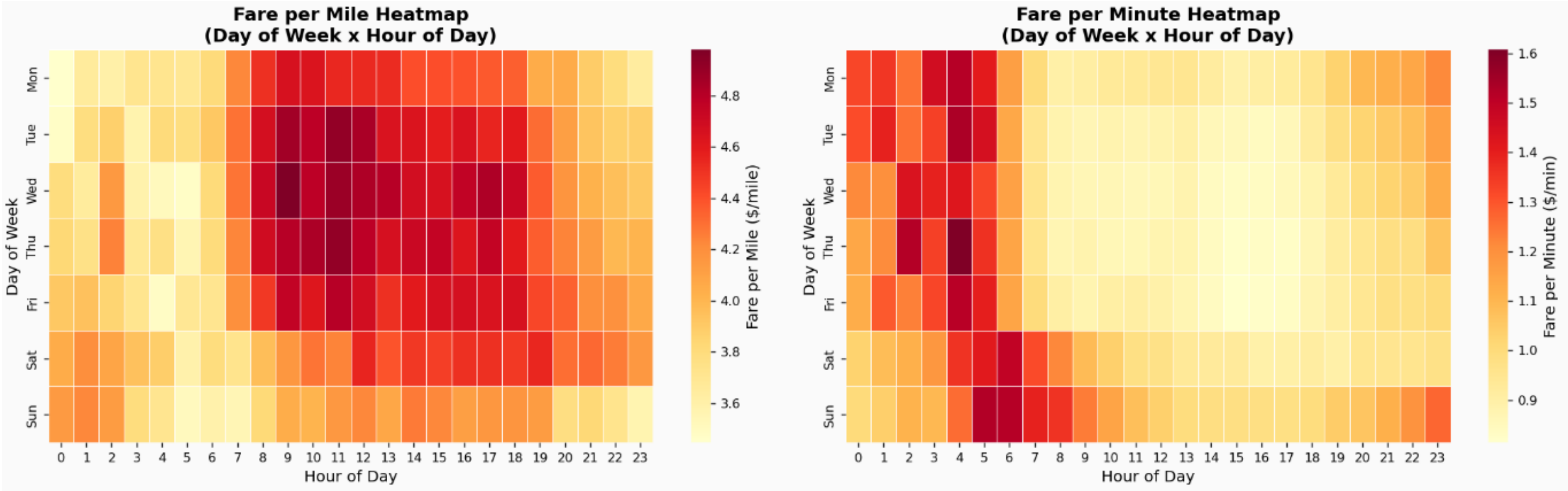
5.7 L2 Fare

- Manhattan zones generally achieve higher fare per mile but lower fare per minute, consistent with shorter, slower trips in congested urban areas.

- Temporal patterns suggest congestion increases fare per mile during peak periods, while off-peak periods tend to produce more time-efficient trips and higher fare per minute.

- The contrasting behavior of these 2 metrics supports a zone- and time-specific operating model, with both KPIs used jointly for operational monitoring and shift planning.





6. CONCLUSIONS

NYC TAXI DATASET – KEY FINDINGS

1 TIME & LOCATION MATTER MOST

- Performance is highly dependent on time and location.
- Citywide averages mask significant zone-and time-level differences.
- Decisions should be informed at the zone and time level.

2 WEEKDAYS LATE AFTERNOONS, AND MANHATTAN DOMINATE DEMAND

- Manhattan accounts for the majority of trips.
- Weekdays see higher demand.
- Late afternoons are the busiest time of day.

3 REVENUE IMPROVED OVER TIME, ESPECIALLY IN 2023

- Average revenue shows a clear upward trend over the years.
- The strongest improvement was observed in 2023.

4 CARD ACCOUNTS FOR OVER 75% OF PAYMENT. MOST TRIPS HAVE 20% TIPPING RATE.

- Card payments dominate transactions.
- Common tipping rate is 20%.

5 EFFICIENCY PER MILE AND PER MINUTE CAN MOVE IN DIFFERENT DIRECTION

- Revenue per mile and revenue per minute do not always move together.
- Management decisions should balance both efficiency metrics.

6 DEMAND IS CONCENTRATED IN TWO VENDORS AND STANDARD-RATE TRIPS

- A small number of vendors account for the majority of trips.
- Standard-rate trips represent the largest share of demand.

7 FARE AND TIPS INCREASE WITH TRIP DISTANCE AND DURATION

- Fare and Tip are higher for trips with longer distance and longer duration.
- Both are different per time and location.

The NYC taxi market is driven by time, location, and trip characteristics, with demand concentrated in Manhattan, card payments dominating transactions, and revenue improving over time. Operational decisions should focus on zone-level performance and balance both revenue-per-mile and revenue-per-minute efficiency metrics.

6. CONCLUSIONS

Thank you!